

Captura de Diagramas Esquemáticos:

Diseño jerárquico, multicanal y conceptos avanzados

Basado en materiales elaborados para Electratraining (www.electratraining.org) por:

Miguel Albero Gil (malbero@ai2.upv.es)

Recursos

[Altium Designer Documentation](#) main page

No hay libros sobre Altium Designer

Encontrarás escasa oferta de cursos de Altium fuera de la propia red de Altium

Afortunadamente, la documentación oficial de Altium es extensa y hay multitud de videos en Youtube

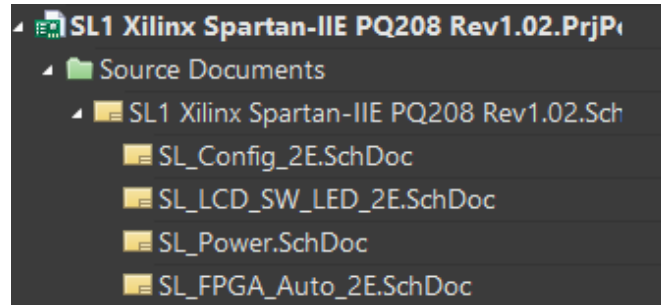


Agenda

- **Hierarchical design**
- SCH panels Adv.
- Device Sheets
- Snippets
- Multichannel

Hierarchical Design

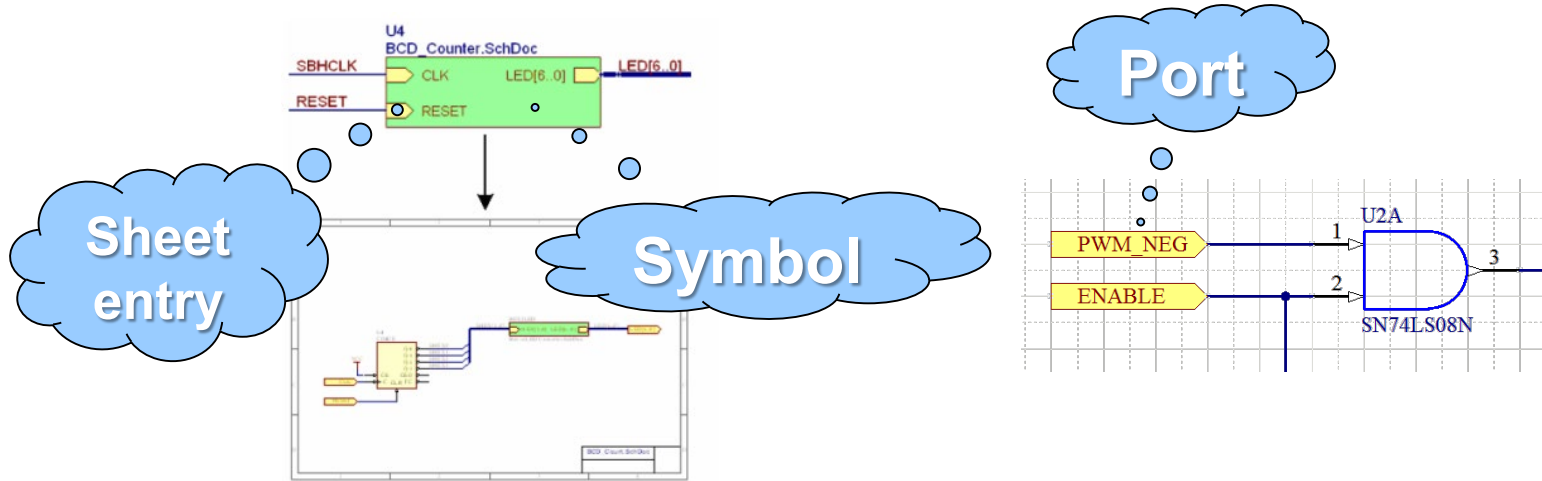
- Altium support multisheet designs with either a flat or hierarchical structure
 - In a **flat design**, all sheets are at the same level
 - In a **hierarchical design**, a sheet instantiates other sheets
- When the project is compiled, the hierarchy is recognized in the Project panel



- FPGA Projects allow mixed hierarchies using both VHDL and schematics

Hierarchical Design: Sheets and Symbols

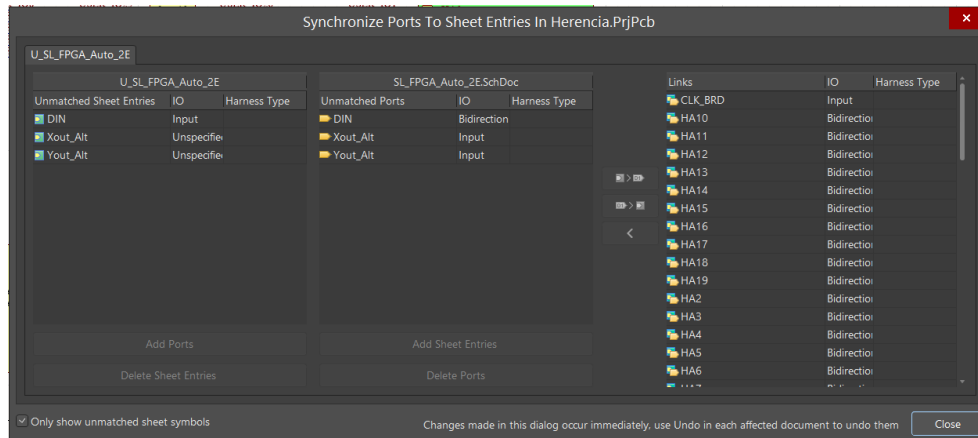
- A **sheet** is instantiated in another sheet with a **symbol**



- Connections between sheets are made through **ports** in the instantiated sheet and **entries** in the symbol

Working with hierarchical Designs

- You can move around hierarchies with the  button (Tools menu)
 - If you press over a symbol, it takes you to the subsheet
 - If you press over a port, it takes you to the top level
- **Design ➔ Synchronize Sheet Entries and Ports**
 - To check if there are mismatches in connectivity

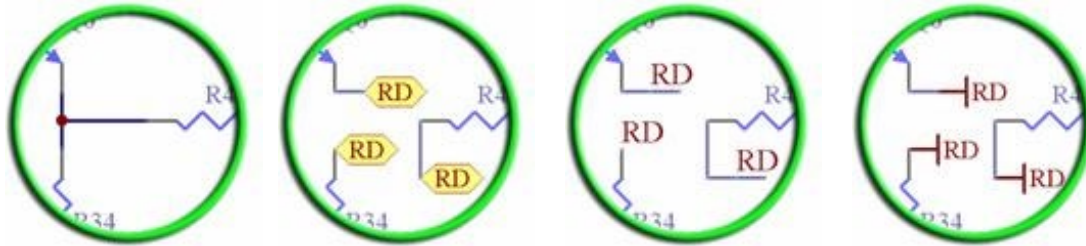


Creating hierarchical Designs

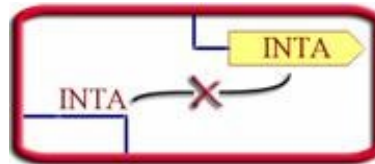
- **Top-down approach:** Draw a sheet symbol, then:
 - Create sheet from symbol
 - Create VHDL/Verilog file from symbol
- **Bottom-up approach:** Draw a sheet, then:
 - Create symbol from sheet or HDL
- Part operations:
 - Push part to symbol
 - Creates a new sheet with just the part, and a symbol on the original sheet that references the new sheet
 - Convert part to sheet symbol
 - Creates a new symbol with the same connections as the part, but it is not associated to any schematic sheet

Net connectivity

- The only way to pass signals between sheets in a project is with **net identifiers**
 - Net identifiers can also be used locally within a single sheet
 - Net labels, ports and power ports are generically known as net identifiers

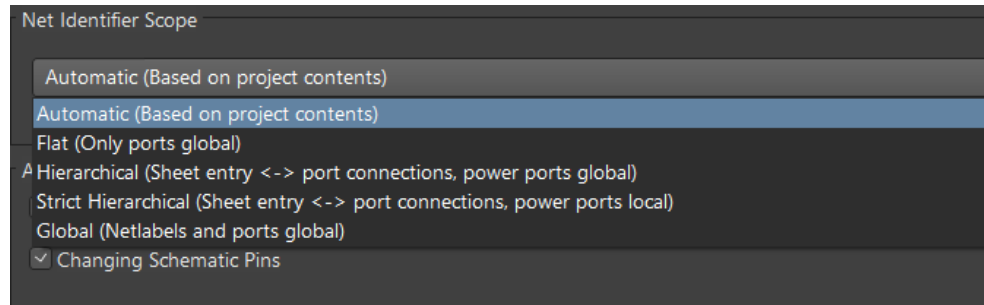


- Net identifiers of different types (such as net labels and ports) will NOT connect even though their names match



Connectivity Scope (in project options)

- **Scope** specifies how you want the net identifiers to connect
 - **Flat**: Ports connect globally between all sheets
 - **Global**: Ports AND net labels connect globally between all sheets
 - **Hierarchical**: Connections are only created vertically between a port and the matching sheet entry
- **Net Identifier Scope** in the **Options** tab of the **Project Options**
 - There is also an **Automatic** option, which is the best option



- Two special net identifier objects that are always global:
 - Power ports and hidden pins

Net identifiers

 NetLabel

- Net label connects:
 - Within a sheet to net labels of the same name
 - Horizontally between sheets if the Net Identifier Scope is set to Global



- Port connects:
 - Vertically if it is matched to a sheet entry on the parent sheet symbol and either the Hierarchical or Automatic Net Identifier Scope is used
 - Horizontally to all matching ports when either the Flat or Global scope is used



- Sheet Entry connects
 - Vertically down to a port on the sheet referenced by the symbol



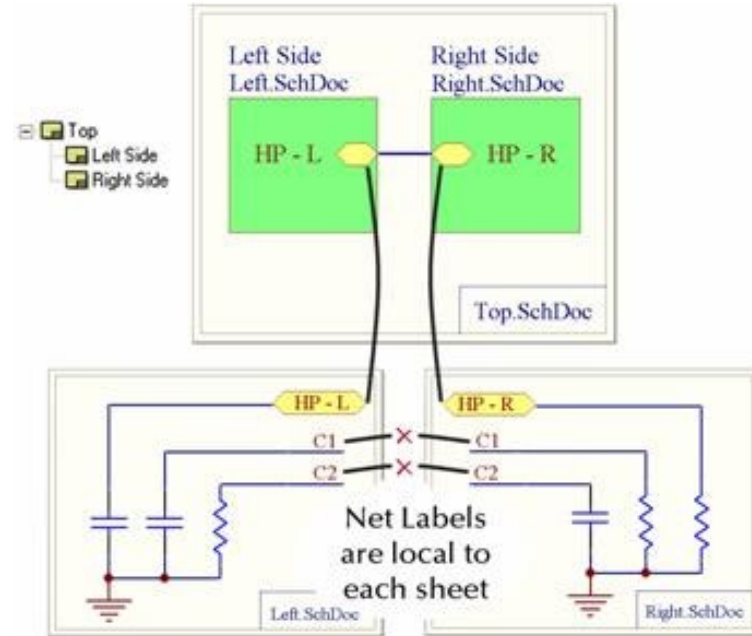
- Off-sheet Connector connects
 - Horizontally to matching off-sheet connectors in flat designs, otherwise they are limited to sheets referenced within a single, "sub-divided" sheet symbol
 - Port Cross-References cannot be applied to off-sheet connectors; therefore, Ports should be used where possible



- Power Port connects
 - Globally to all matching power ports across your entire project
- Hidden Pin connects
 - Globally to all hidden pins in your project that have a matching Connect To value

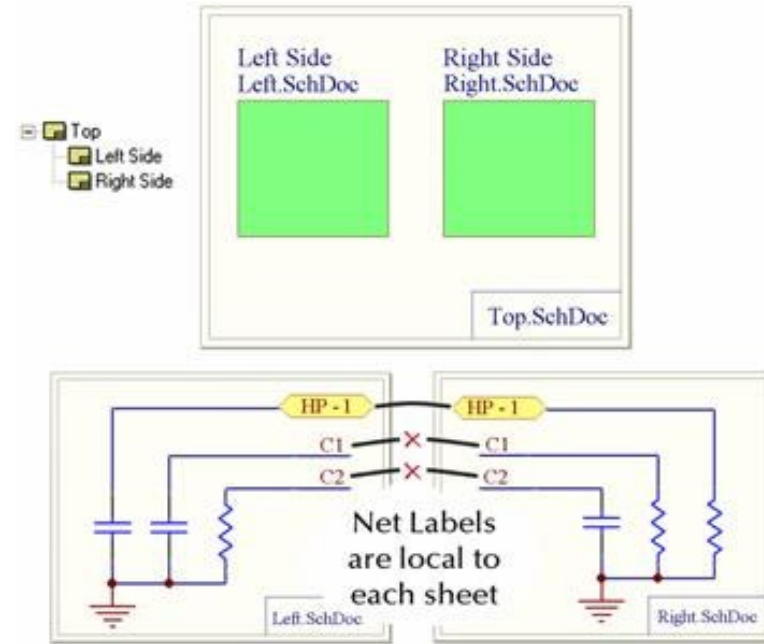
Connectivity example: Hierarchical

- Project will **automatically** be detected for hierarchical scope
 - Sheet entries are present within the sheet symbols on the parent sheet
 - Net labels C1 and C2 on each sub-sheet will not connect to matching net labels on the other sub-sheet
 - Even if ports had same names, they would not make horizontal connections with one another between sub-sheets when the hierarchical scope is on



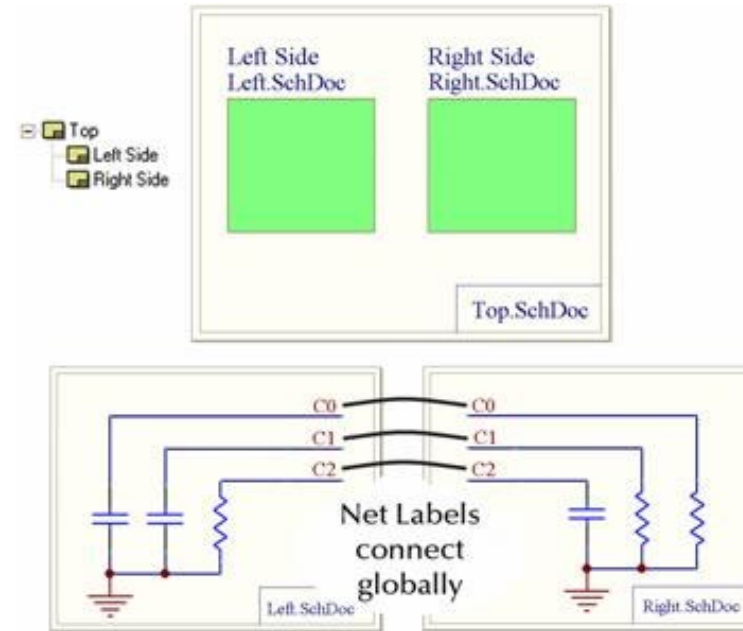
Connectivity example: Flat (ports global)

- Project will **automatically** be detected for only ports global scope
 - The absence of any sheet entries, causes the scope to automatically change to ports global
 - Project is flattened as far as ports are concerned
 - As the project is flat, the top-sheet could be removed from the project and the design would still compile correctly



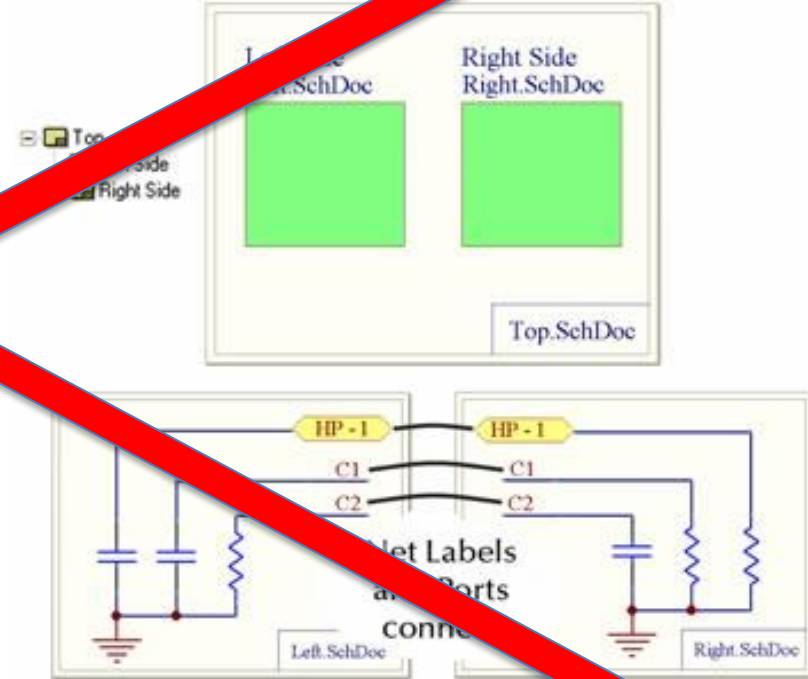
Connectivity example: Global

- Project is entirely devoid of both sheet entries and ports.
 - The only case in which net labels will automatically make global connections in a multi-sheet design
 - Net labels will connect with all matching net labels in the project, regardless of the structure
 - As this design project is flat, the top-sheet could be removed from the project and the design would still compile correctly



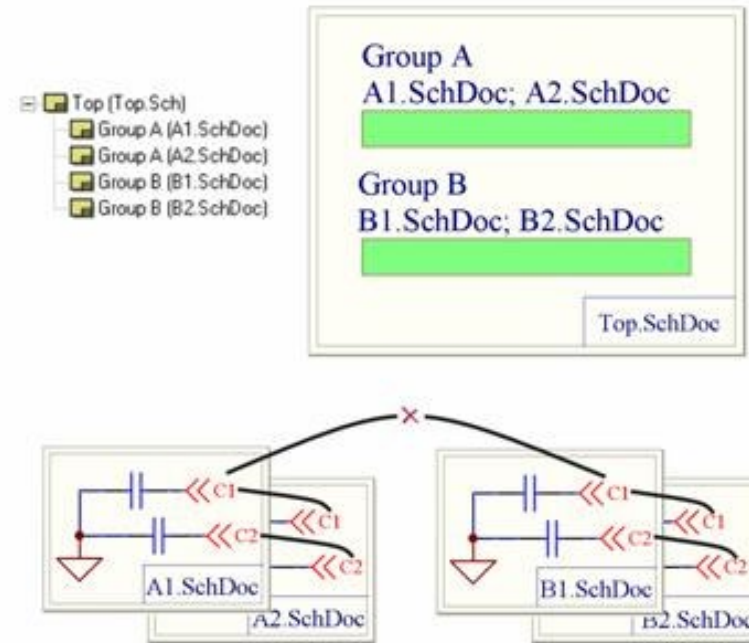
Connectivity example: Net Labels and Ports Global

- Legacy option, must be explicitly selected
 - It is not automatically detected
 - As the project is flat, the top-sheet could be removed from the project and the design would still compile correctly



Connectivity example: Off-Sheet

- Top sheet references 4 sub-sheets with only 2 sheet symbols.
 - Multiple schematic files are referenced in the sheet symbol's Filename field, separated by a semicolon
 - Off sheet connectors will then create flat connections between all sheets grouped within a single sheet symbol, even if other sheets in the project are using vertical connectivity
 - Many designers may never use off-sheet connectors. They were introduced as a means of importing designs seamlessly from other design packages

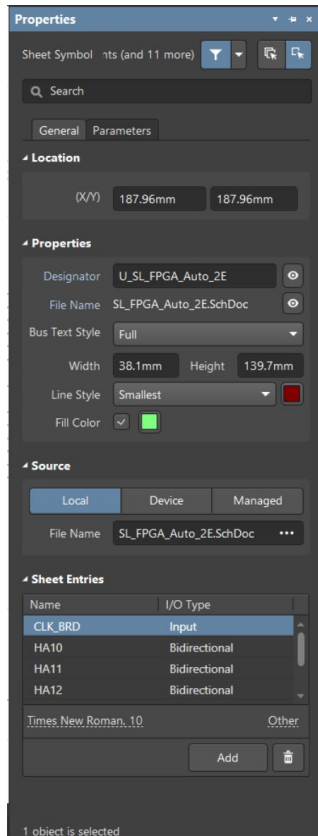




Agenda

- Hierarchical design
- **SCH panels Adv.**
- Device Sheets
- Snippets
- Multichannel

SCH Properties and List Panel



- Complementary panels
- **Properties** to see and change a parameter in one object
- **SCH List** to view parameters and change them in one or more objects

The SCH List panel displays a table of objects from the current document. The table includes columns for Object Kind, X1, Y1, Orientation, Description, Locked, Mirrored, Lock Designator, Lock Part ID, Pins Locked, File Name, Configuration, and Library. The table lists various components such as Headers, Surface Mount Crystal Oscillators, Capacitors, Resistors, Accelerometers, and Jumpers.

Object Kind	X1	Y1	Orientation	Description	Locked	Mirrored	Lock Designator	Lock Part ID	Pins Locked	File Name	Configuration	Library
Part	3100mil	7700mil	0 Degrees	Header, 10-Pin, Dual row	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	5100mil	7700mil	0 Degrees	Header, 10-Pin, Dual row	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	10300mil	2400mil	0 Degrees	Surface Mount Crystal Oscillator	☐	☒	☐	☐	☒	*	*	SpiritLevel_2i
Part	10000mil	3000mil	90 Degrees	Capacitor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	12300mil	4600mil	0 Degrees	Header, 5-Pin, Dual row	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	3800mil	3700mil	90 Degrees	Capacitor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	2900mil	4700mil	0 Degrees	Capacitor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	2700mil	4700mil	0 Degrees	Capacitor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	2400mil	4500mil	90 Degrees	Resistor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	9300mil	6300mil	90 Degrees	Jumper, 2-Pin	☐	☒	☐	☐	☒	*	*	SpiritLevel_2i
Part	9300mil	5900mil	180 Degrees	Resistor	☐	☒	☐	☐	☒	*	*	SpiritLevel_2i
Part	3600mil	5200mil	0 Degrees	Accelerometer with Digital Output	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	3200mil	5500mil	0 Degrees	Resistor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	5200mil	4900mil	180 Degrees	Resistor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	5200mil	4700mil	180 Degrees	Resistor	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	5400mil	5000mil	0 Degrees	Jumper, 2-Pin	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	5400mil	4800mil	0 Degrees	Jumper, 2-Pin	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	5400mil	5050mil	0 Degrees	Shorting Link, 2.54mm (0.1") Pitch	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	5400mil	4850mil	0 Degrees	Shorting Link, 2.54mm (0.1") Pitch	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i
Part	9150mil	2500mil	270 Degrees	Jumper, 2-Pin	☐	☐	☐	☐	☒	*	*	SpiritLevel_2i

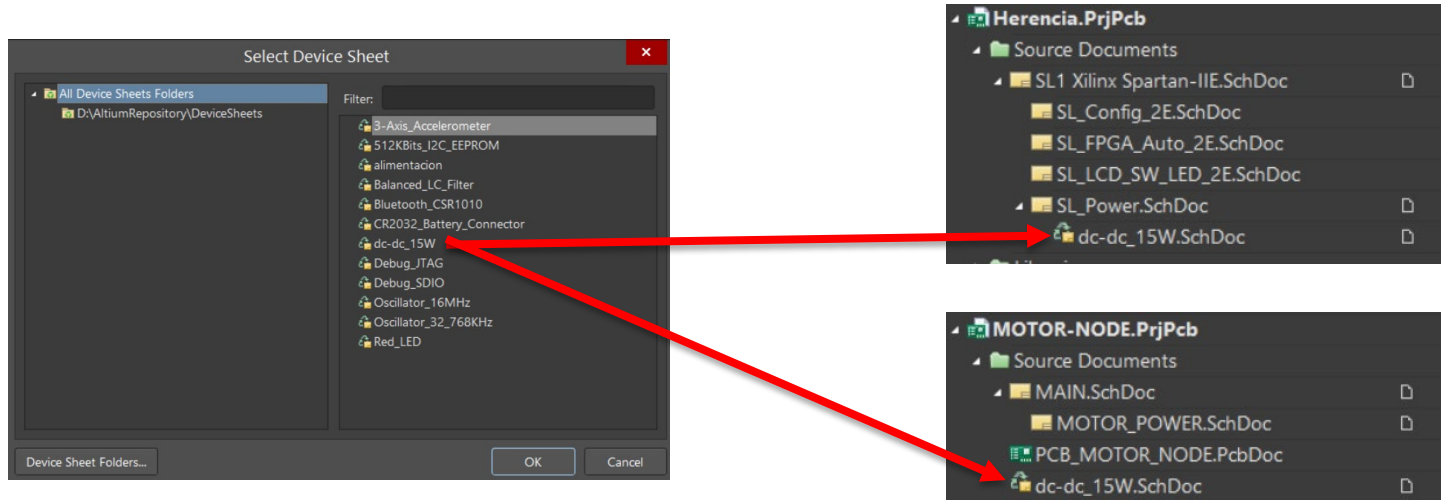


Agenda

- Hierarchical design
- SCH panels Adv.
- **Device Sheets**
- Snippets
- Multichannel

Device Sheets: Concept

- Device Sheets simplify the design process by providing modularized and consistent building blocks which can be re-used between projects
 - Device Sheet Symbols are placed and referenced similarly to components
 - They function in the same way as Sheet Symbols and Schematic Documents but are not explicitly added to projects

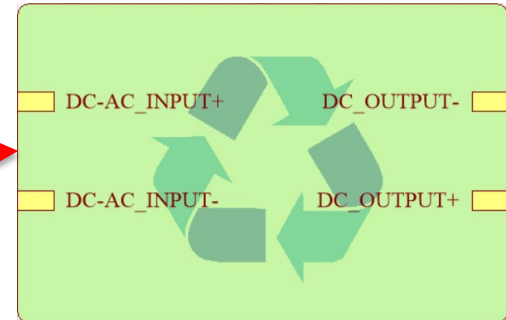
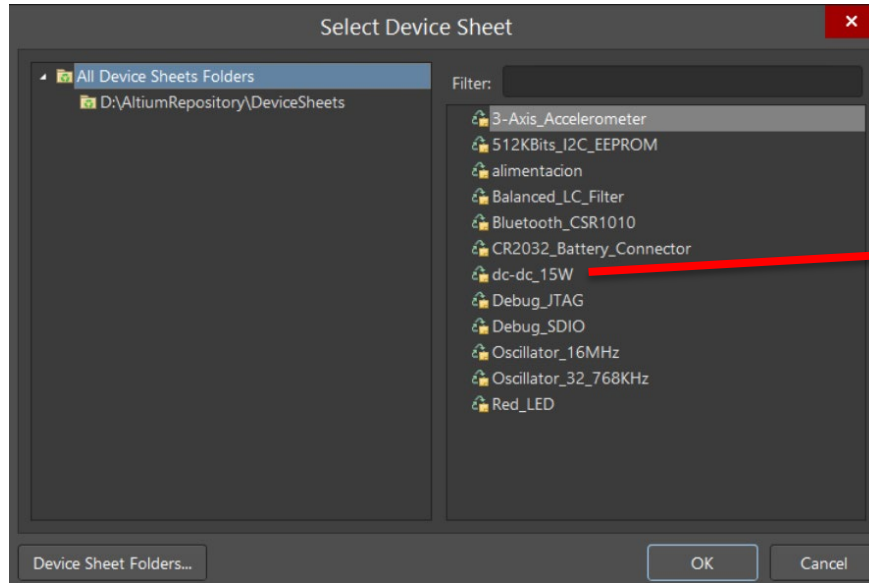


Creating Device Sheets

- A Device Sheet can be any normal Schematic Document including Schematic Documents with Sheet Symbols
 - By default, Device Sheets are **read-only** when they are placed in your project
- Create and save a folder intended to hold your Device Sheets
 - You can create a main folder called Device Sheet Folders with sub folders such as, Data Conversion, Power Management and Interface depending on your requirements
- Create a new Schematic Document or save an existing Schematic Document to the relevant Device Sheet Folder
- Declare the Device Sheet Folder in Altium Designer, in **Data Management**
 - ➔ **Device Sheets**
 - Make sure include subdirectories is enabled

Placing Device Sheets

- Place ➔ Device Sheet Symbol or 





Agenda

- Hierarchical design
- SCH panels Adv.
- Device Sheets
- **Snippets**
- Multichannel

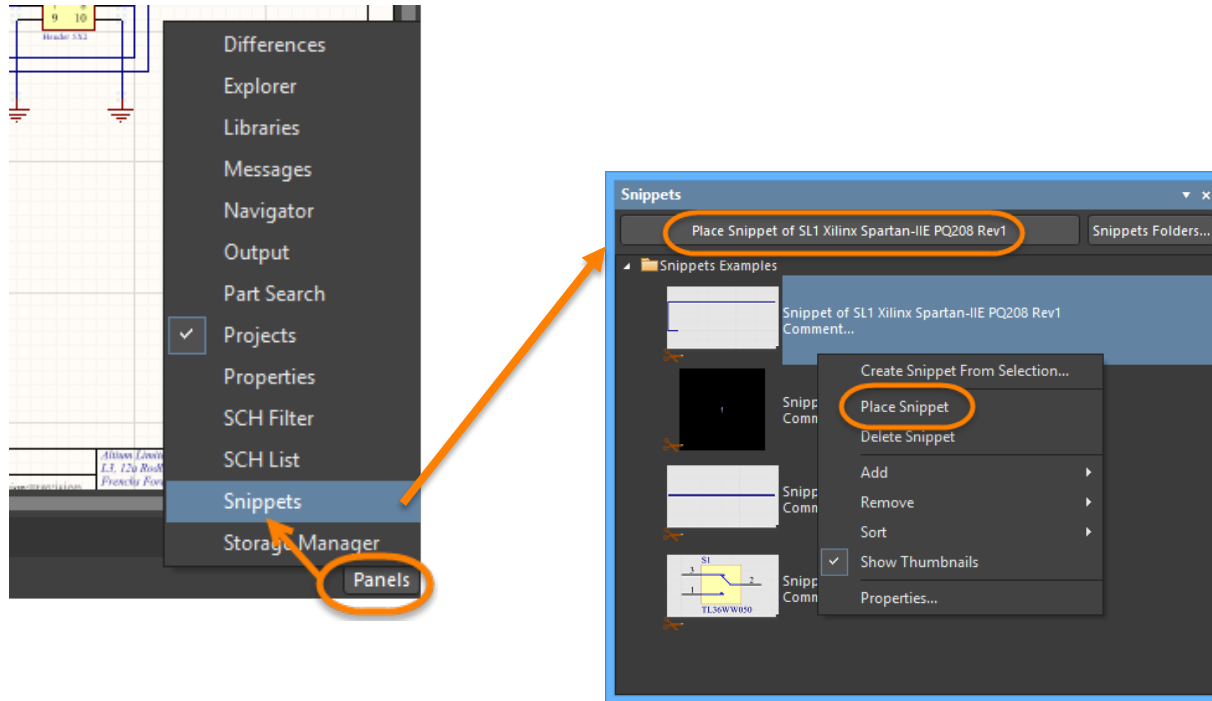
Snippets

- If your designs often include common 'sections' of circuitry or code, then you will make good use of the **Design Snippets** feature
- A simple and easy to use feature, the Snippets system lets you save any selection of:
 - Circuitry on a single schematic sheet
 - Circuitry in a PCB design, including the components and the routing
 - Code from within one of Altium Designer's code-aware editors
- A circuit 'snippet' can be added into any design, without you having to start from scratch each time – **design reuse, made simple**

To Summarize: Copy & Paste using a folder in your disk

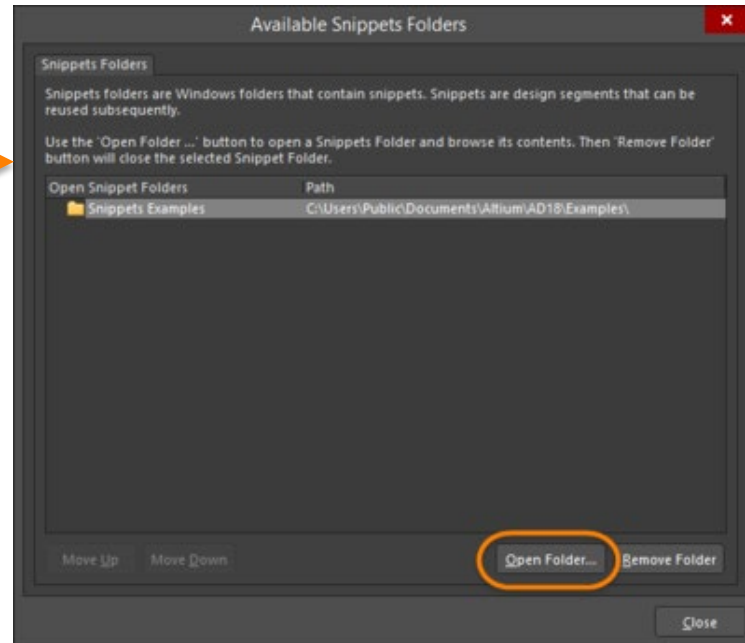
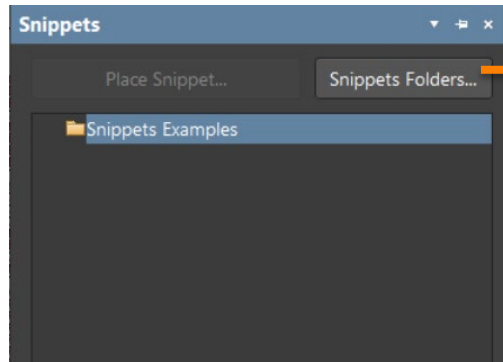
Snippets Panel

- Snippets are accessed through the snippet panel



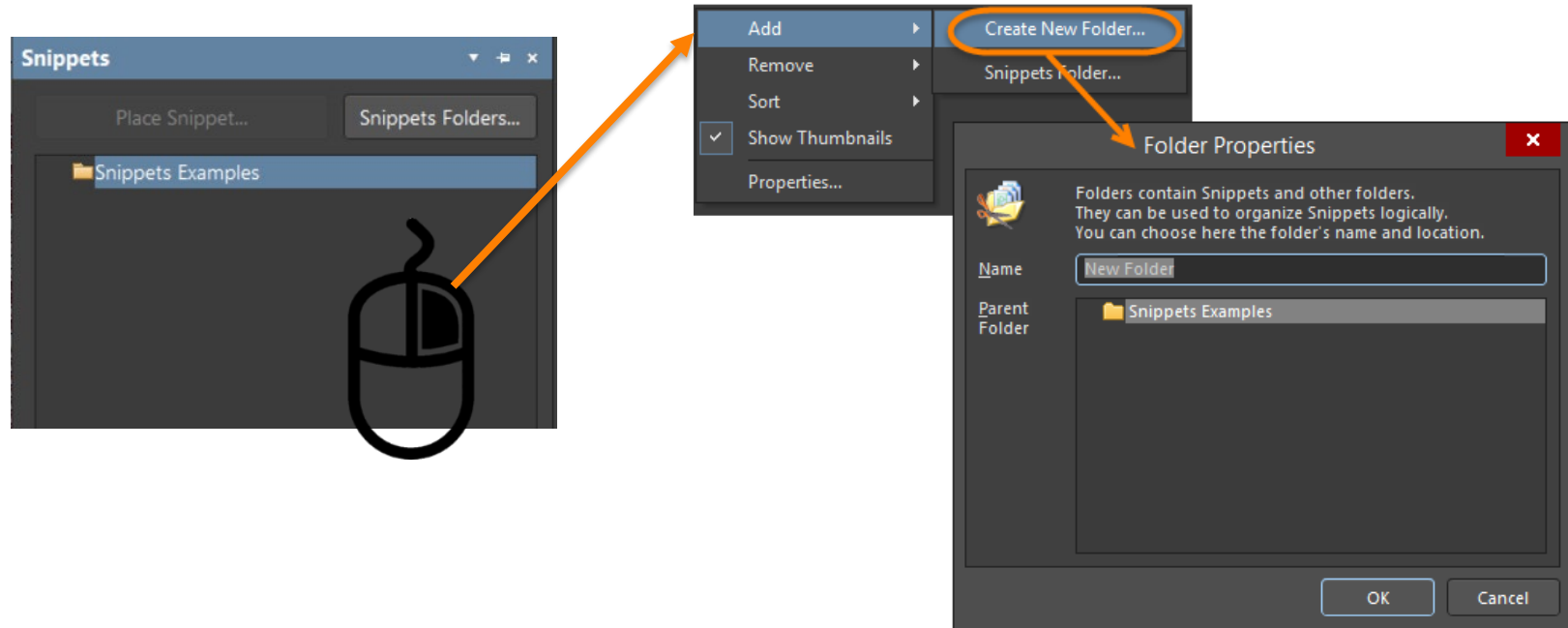
Snippet Folder

- Snippets are stored in folders, that can be accessed using the **Snippets Folder**
- We can manage all our folders



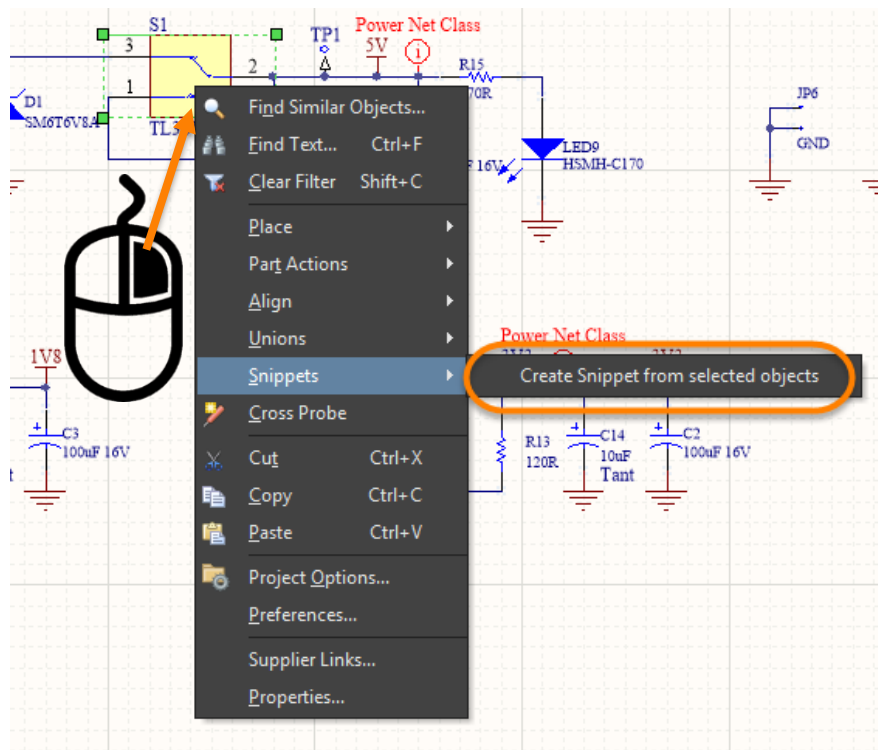
Snippet Folder

- Snippets are stored in folders and **subfolders**. Once we have a Snippet Folder, we can create as many subfolders as we wish.



Creating a Snippet

- Snippets are created from selected objects



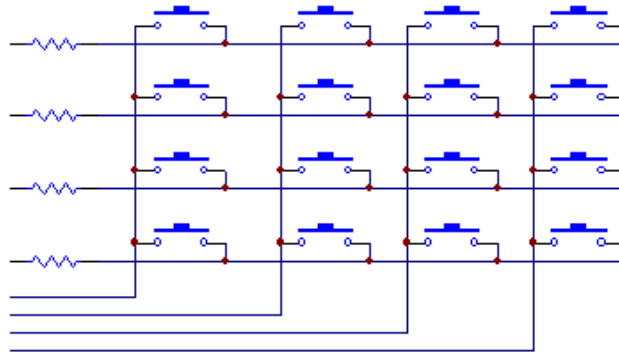


Agenda

- Hierarchical design
- SCH panels Adv.
- Device Sheets
- Snippets
- **Multichannel**

MultiChannel: Concept

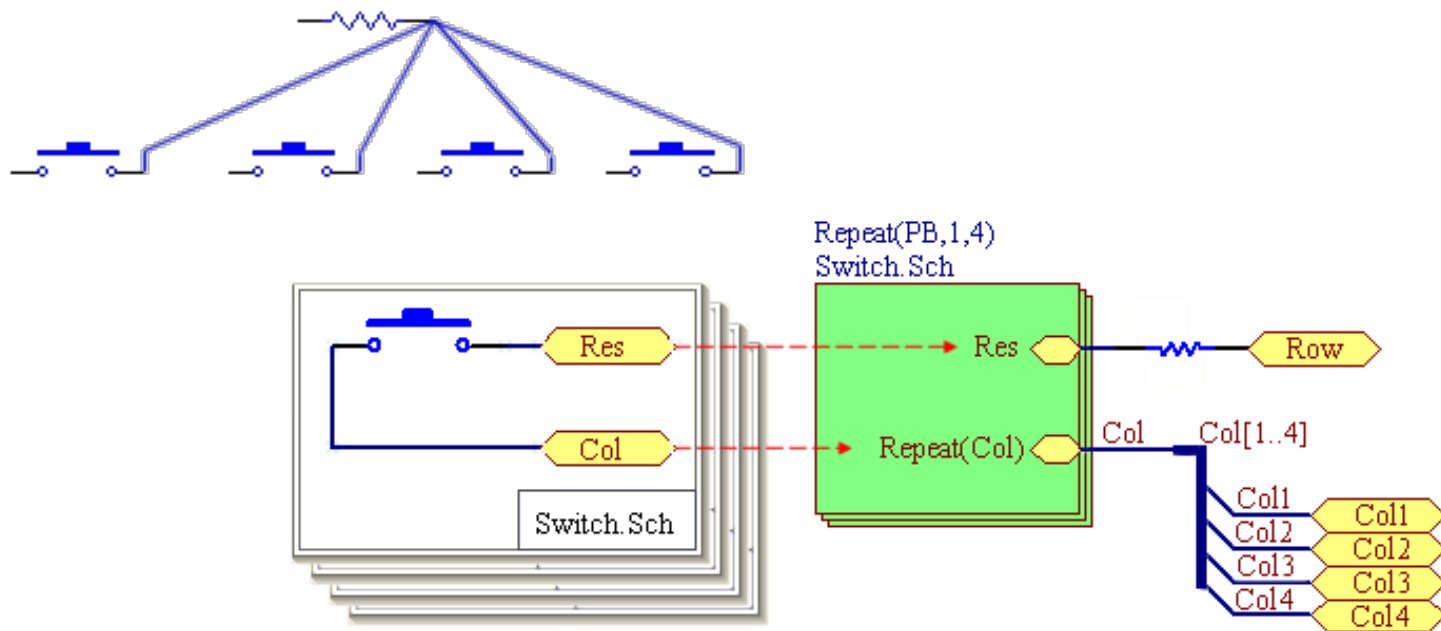
- Many designs contain **repeated circuitry**
 - One board might duplicate the same section thirty-two times, or perhaps contain four identical banks with eight sub-channels each



- Altium Designer offers true multi-channel design, meaning that you can reference single sheets in your project repeatedly
 - It supports **channels nested within other channels**

Multichannel Connectivity

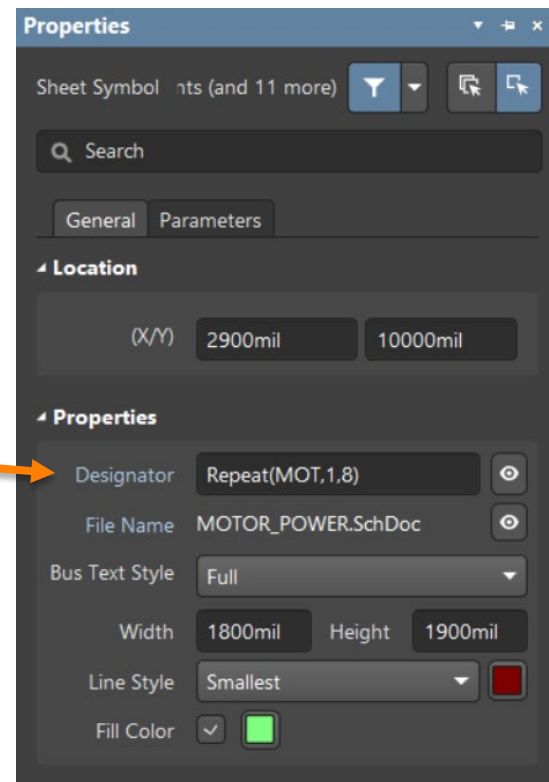
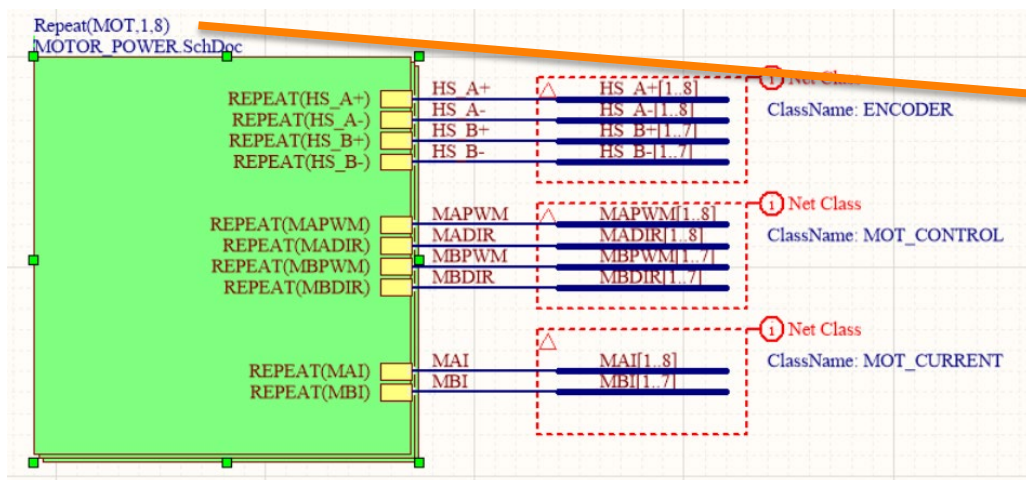
There are two types of sheet-to-sheet connectivity available in multi-channel design: nets that are common to all channels, and nets that are unique to each one



How To Instantiate Multichannel

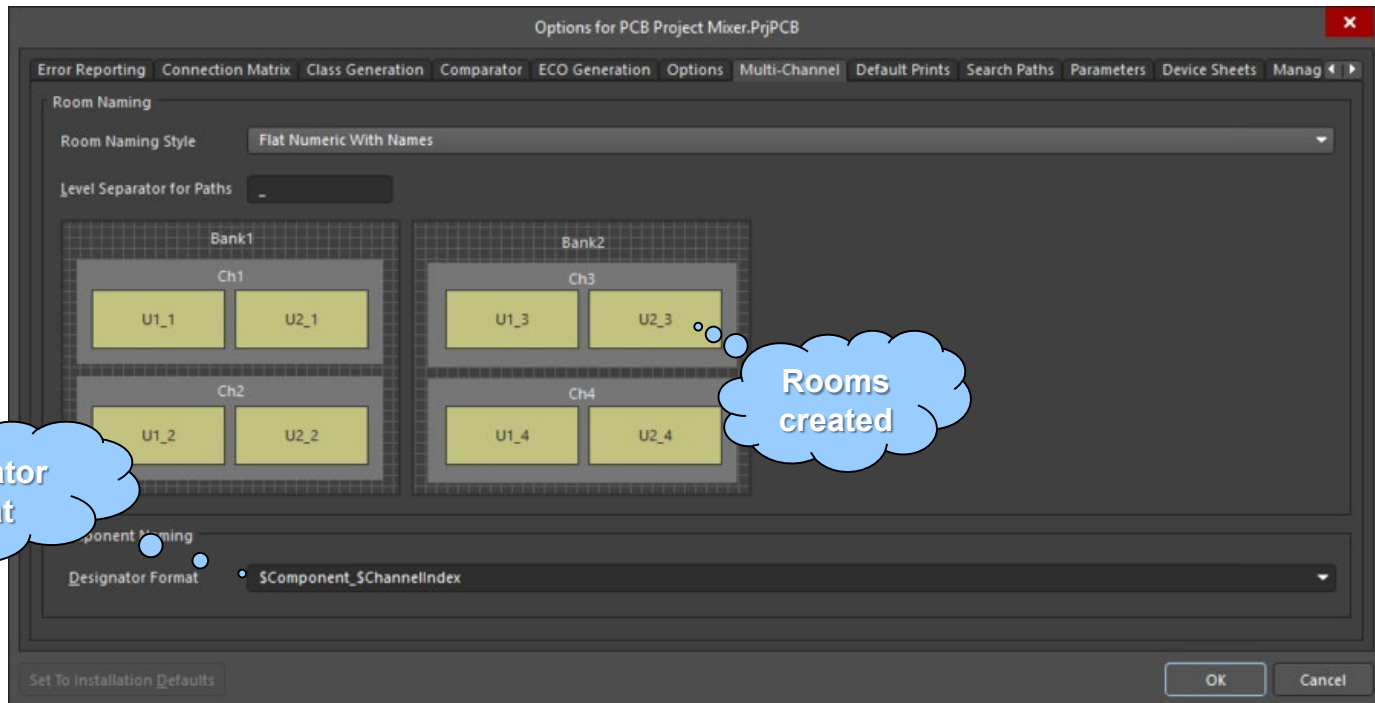
Same as instantiating sheets symbols, but name is

– REPEAT(*Designator*, *Start*, *End*)



Multichannel Options

- Project Options



Multichannel: Component Naming

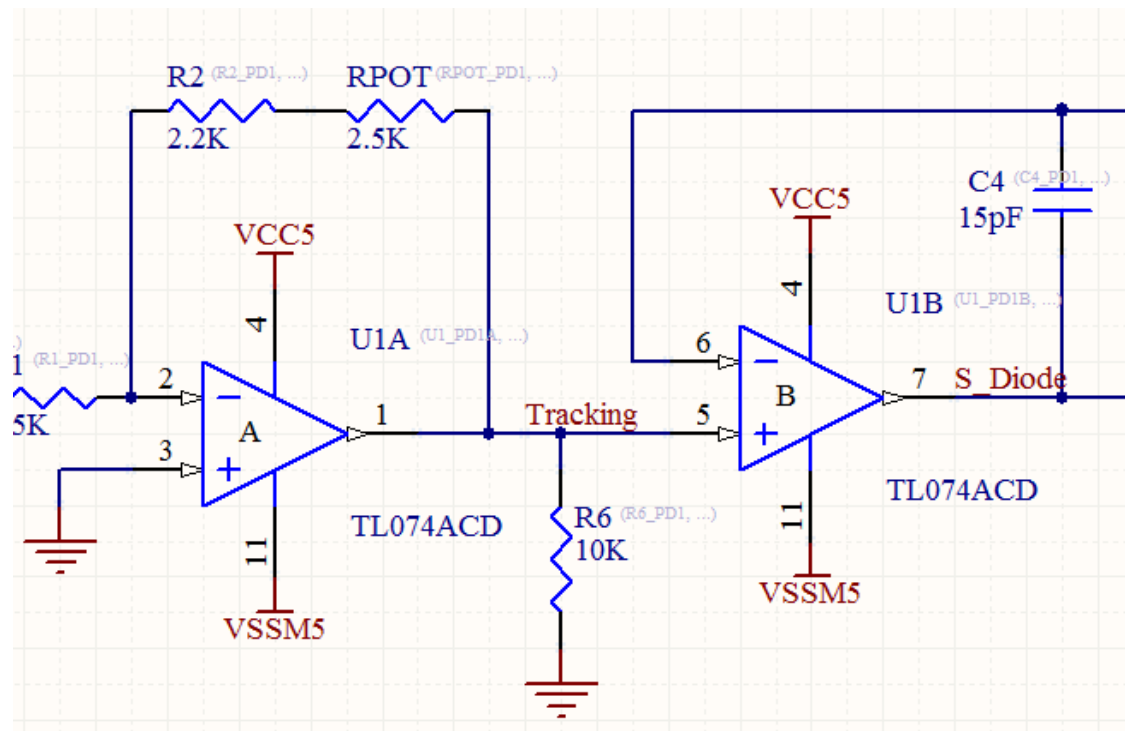
- Eight predefined formats:

Flat designator formats
\$Component\$ChannelAlpha
\$Component_\$ChannelPrefix\$ChannelAlpha
\$Component_\$ChannelIndex
\$Component_\$ChannelPrefix\$ChannelIndex
\$ComponentPrefix_\$ChannelIndex_\$ComponentIndex
Hierarchical designator formats
\$Component_\$RoomName
\$RoomName_\$Component
\$ComponentPrefix_\$RoomName_\$ComponentIndex

- Or custom format

Component Names

Actual names appear grayed out:





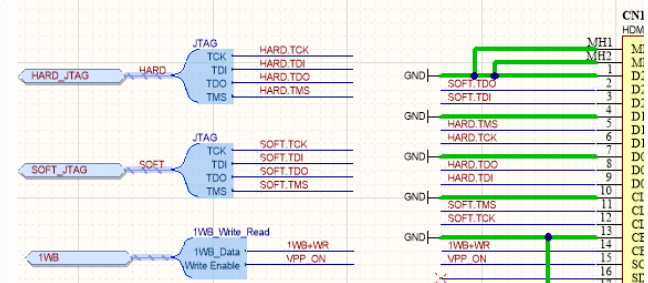
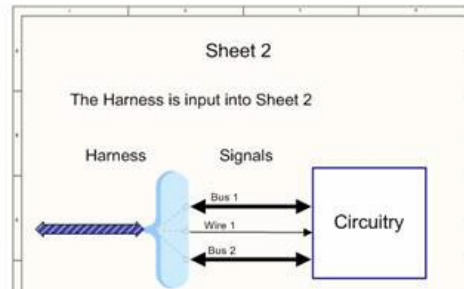
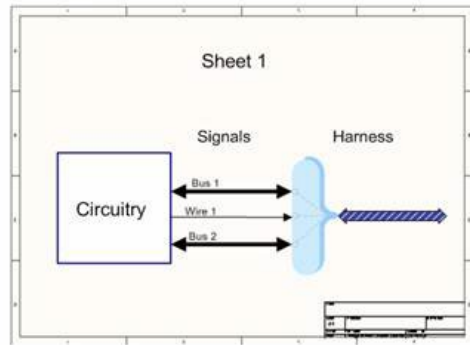
Agenda

- **Harness**
- Supplier search
- Variants
- SCH pre-layout directives
- Pin Swapping

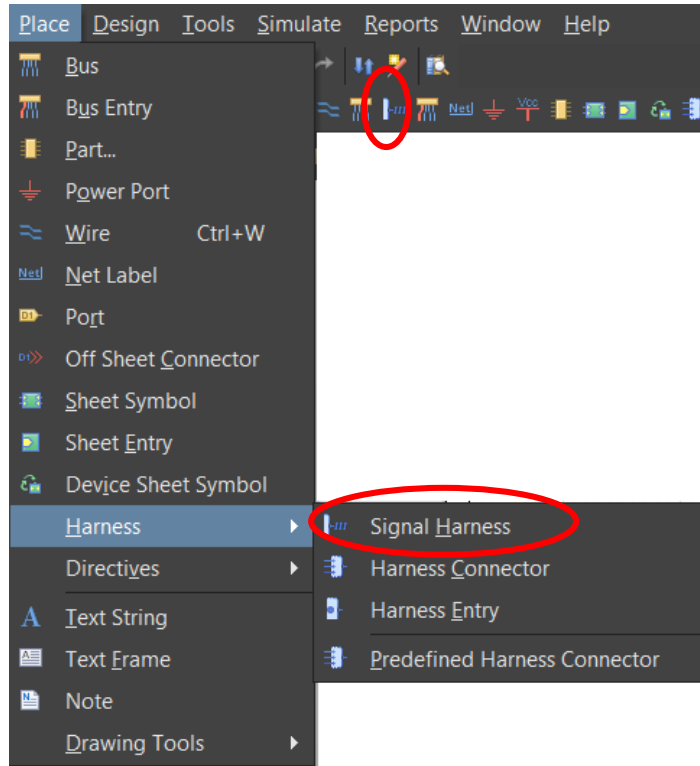
Harness signal connection

- Signal Harnesses allow for a more complex design in the same schematic workspace, increasing readability of designs and building potential for re-use
- There are 4 key elements to a Signal Harness system, accessible through the Schematic Editor:

- Signal Harness
- Harness Connector
- Harness Entry
- Harness Definition File

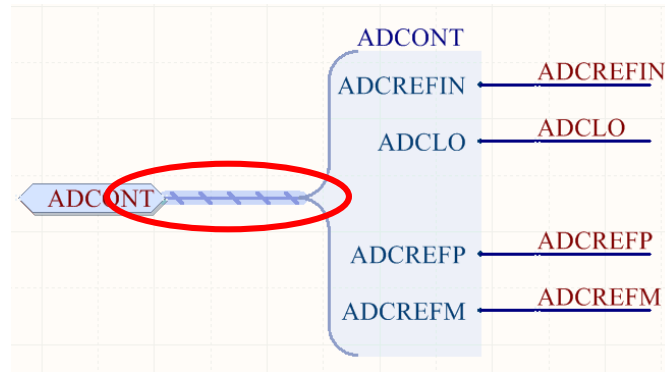


Signal harness

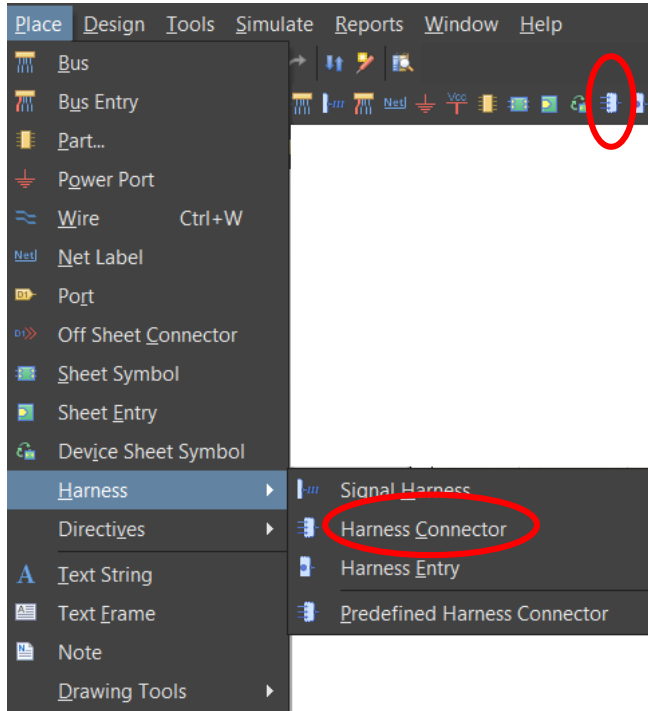


This object represents the abstract connection which combines different signals.

You can use this object to connect different subsystems across your design.

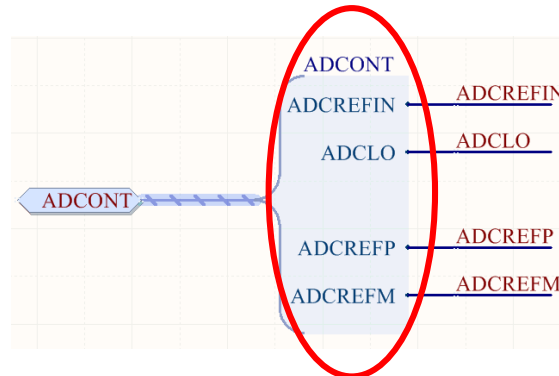


Harness connector

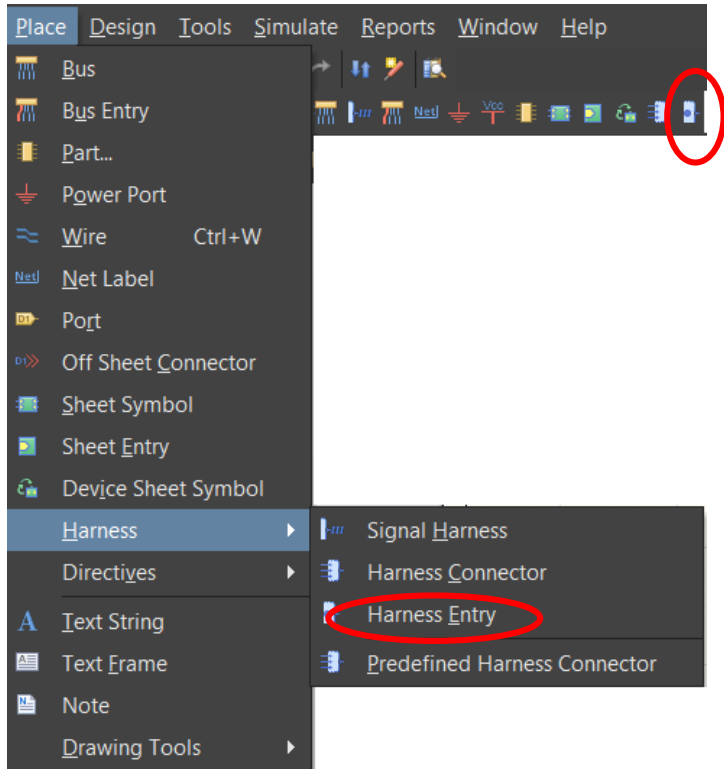


This object is used to group several signals together in order to form a Signal Harness.

It is both a graphical definition of the Signal Harness and a graphical means to include actual nets, buses and other harnesses to the main Signal Harness.

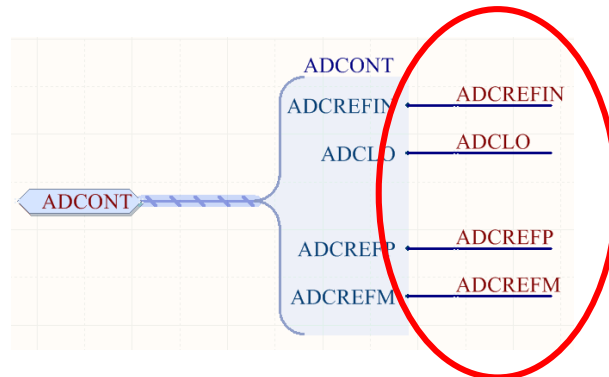


Harness entry

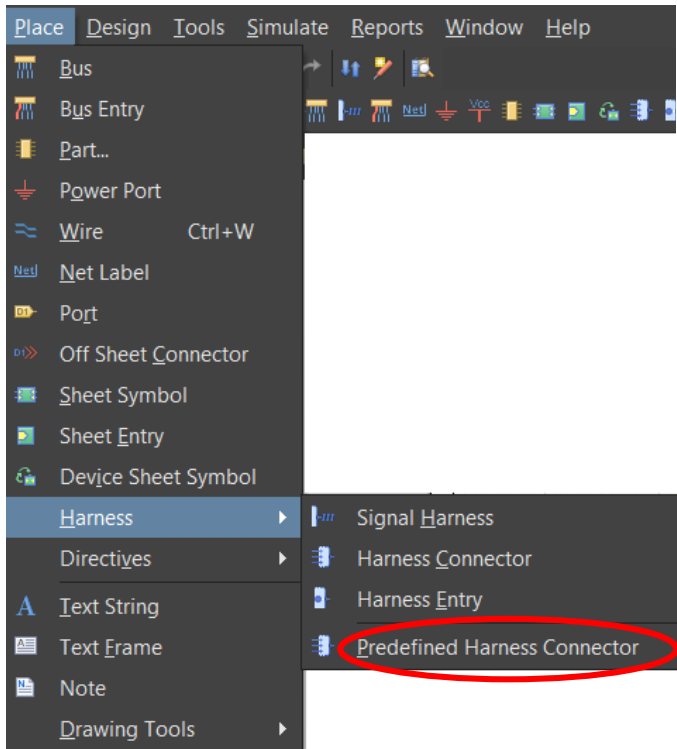


A Harness Entry is the graphical definition of a Signal Harness member.

It is also the connection point through which actual nets, buses and Signal Harnesses are combined to form a higher level Signal Harness.

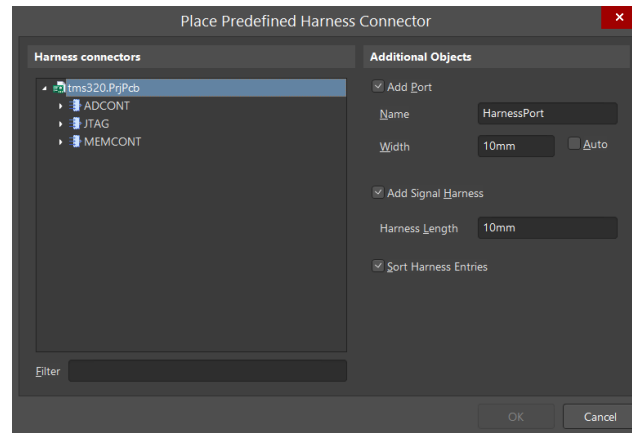


Predefined harness connector



Predefined Harness Connector dialog appears with a list of all available Harness Connectors in your current and open projects and in your Sheet folder.

In this dialog, you can choose to add and name a port of specified width and to add a Signal Harness of specified width.

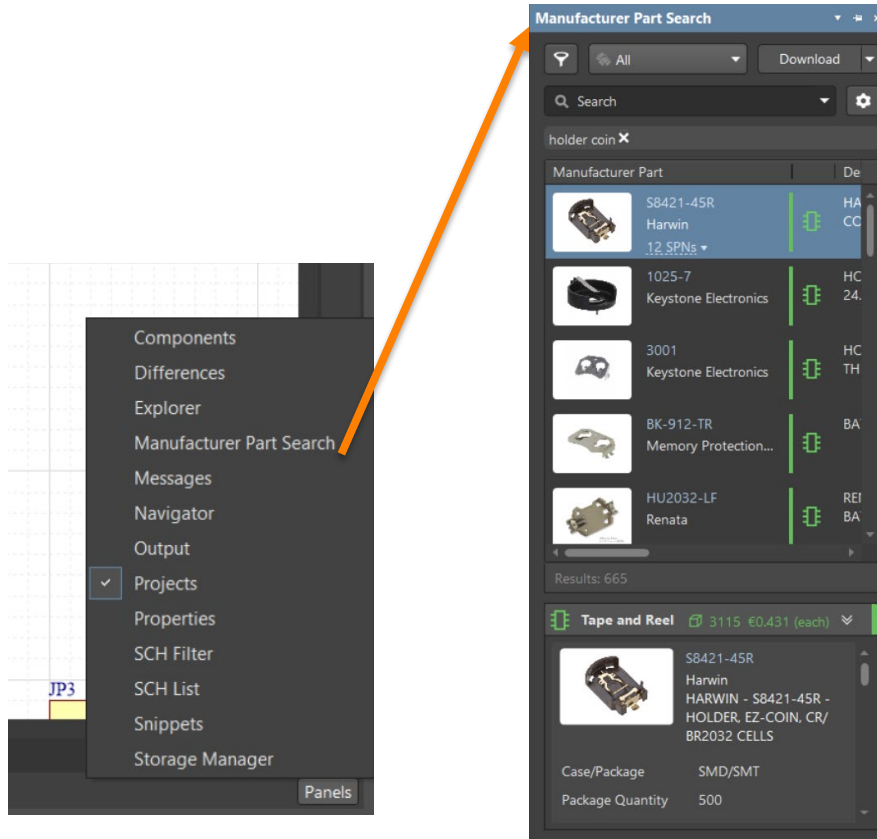




Agenda

- Harness
- **Supplier search**
- Variants
- SCH pre-layout directives
- Pin Swapping

Manufacturer PART SEARCH PANNEL

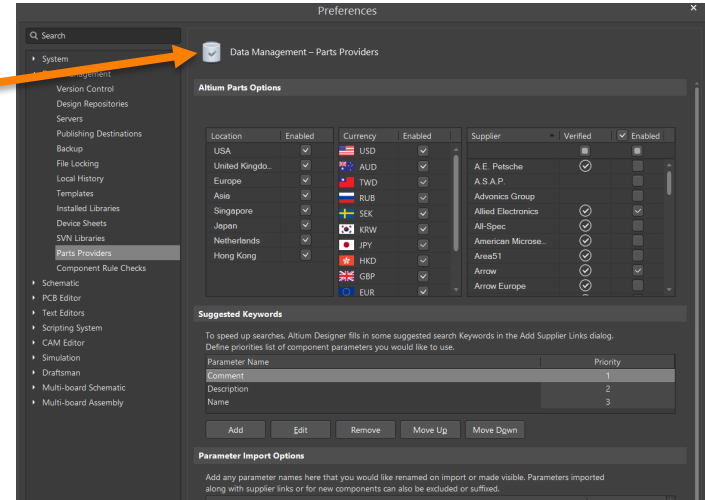
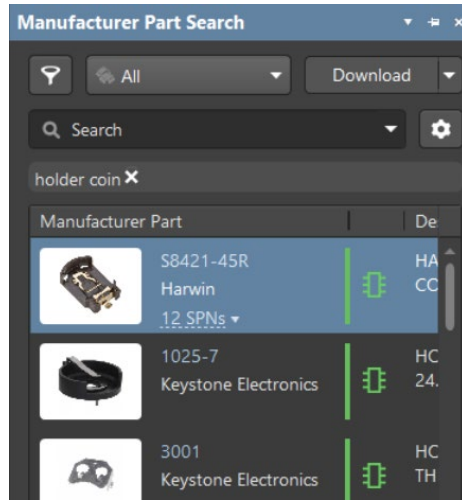


Through direct connection to supplier web services, you are able to search across a supported supplier's entire product catalog.

This 'live' data can then be integrated within your design process.

In short, this feature allows you to search supplier databases from within Altium Designer and link your design components to matching Supplier Items.

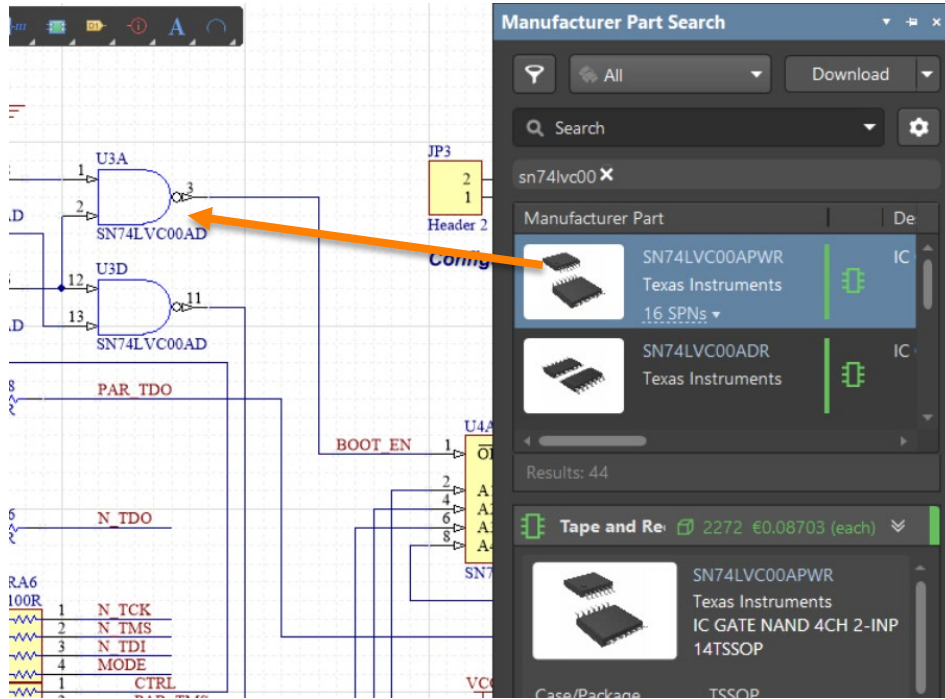
Manufacturer PART SEARCH PANNEL



The Manufacturer *Part Search* panel provides the interface to the live supplier data. It is a direct portal to parts databases provided through supplier web services.

Enter one or more keywords of the part you want to search for in the **Keywords** field. Click **...** to the right of the **Keywords** field to access the **Search Filters** pop-up.

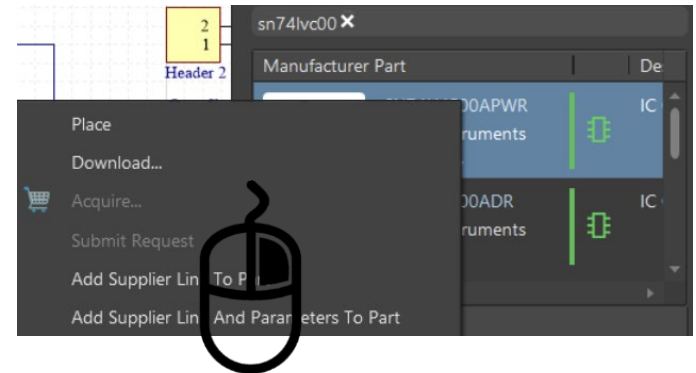
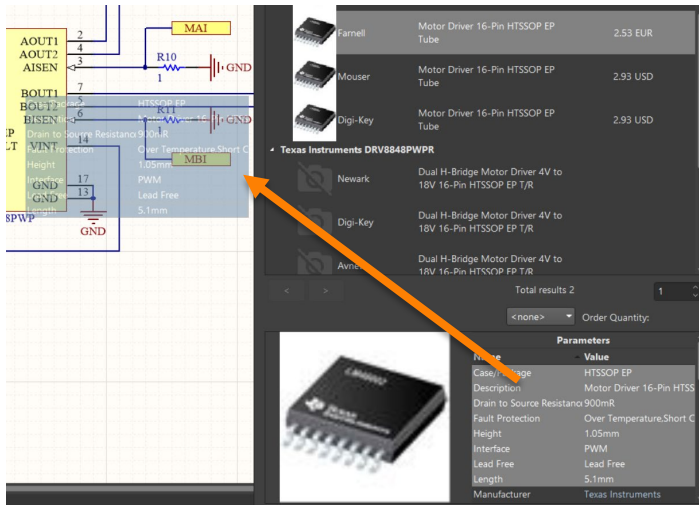
Manufacturer PART SEARCH PANNEL



Creating a 'Supplier Link' is a simple, straightforward process. Simply click on the required Supplier Item, in the results region of the **Supplier Search** panel, then drag and drop onto the target component.

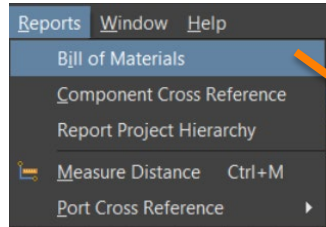
Manufacturer PART SEARCH PANNEL

For any given Supplier Item found through a search in the **Supplier Search** panel, you can import its parameters, data sheet links, pricing and stock information, as parameters of a target library component (SchLib, DbLib, SVNDbLib), or placed component on a schematics sheet.



BILL OF MATERIALS

One of the most important output list generators for PCB designers is the bill of materials, because it represents the main list of devices for the project.



Bill of Materials for Project [BOC1_06.PjPcb] (No PCB Document Selected)

Comment	Description	Designator	Footprint	LibRef	Quantity
1 10uF	Polar Capacitor	C1, C5, C9, C22, C...	MCCT-B	CAP_POLAR	14
2 100nF	Capacitor	C2, C3, C4, C6, C7...	CC2012-0805	CAP	40
3 470uF 16V	Polar Capacitor	C15, C16, C21, C24	R88/3.5	CAP_POLAR	4
4 22nF	Capacitor	C18, C36	CC2012-0805	CAP	2
5 Do Not Install	Capacitor	C28, C31	CC2012-0805	CAP	2
6 470pF	Capacitor	C40	CC2012-0805	CAP	1
7 4700pF	Capacitor	C41, C44	CC2012-0805	CAP	2
8 1uF	Capacitor	C43, C45	C3216-1206	CAP	2
9 220pF	Capacitor	C50, C51, C52, C53	CC2012-0805	CAP	4
10 100pF	Capacitor	C54, C55, C56, C5...	CC2012-0805	CAP	5
11 1nF	Capacitor	C77, C78, C79, C8...	CC2012-0805	CAP	12
12 QC49/SMD	SMD Crystal 20M...	CR1	QC49/SMD	QC49/SMD	1
13 BAS40-04	Shottky Diode Pair	D1, D2, D3, D4, D5	SO-G3	BAS40-04	5
14 BAS40	Shottky Diode	D6	SO-G3	BAS40	1
15 SMBJ53418	Zener Diode, 6.2V...	D7	DO214	BZG03C13	1
16 Header 2	Header, 2-Pin	G1, G2, G3, G4...	HDR1X2	Header 2	9
17 Header 5X2	Header, 5-Pin, Du...	HDR1, HDR2, HD...	HDR2X5	Header 5X2	4
18 70246-2622	Header, 13-Pin, D...	HDR3	HDR2X13_CEN_LP	Header 13X2A	1
19 70246-1022	Header, 5-Pin, Du...	HDR6, HDR7	HDR2X5_LP	Header 5X2	2
20 Header 7X2	Header, 7-Pin, Du...	HDR8	HDR2X7	Header 7X2	1
21 Header 10X2	Header, 10-Pin, D...	HDR9, HDR10	HDR2X10	Header 10X2	2
22 D89 Female	Receptacle Assem...	J1	DSUB1385-2H9	D Connector 9	1
23 D Connector 9_M...	Receptacle Assem...	J2	DSUB19_3MALE	D Connector 9_M...	1
24 Dsub-15-Female...	Dsub-15 High den...	J3	DB15-VGA-Female	Dsub-15-Female...	1
25 8918-P6	6 Pin Through Hol...	J4, J5	PS2-6PIN	PS2-6PIN	2
26 K1 D_0707_8	Low Voltage Group	K6, K7	K1 D_0707	K1 D_0707_8	2

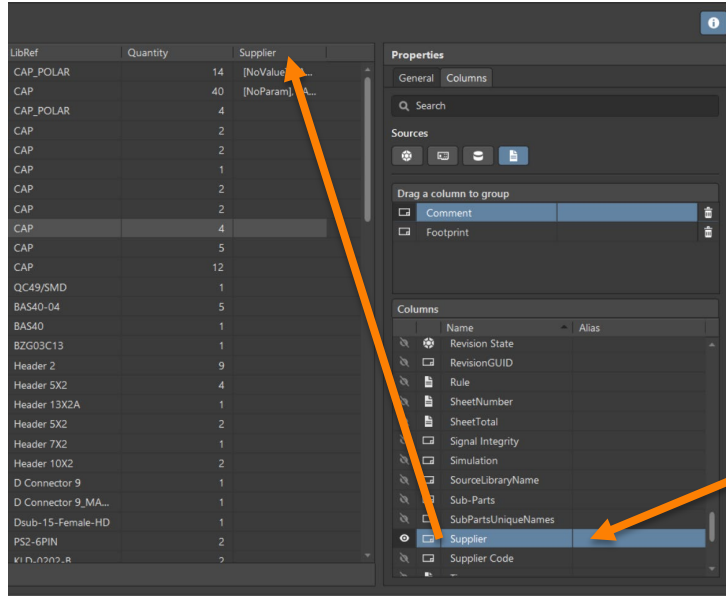
83 of 83 lines visible

Properties: General Columns
Search
Sources
Drag a column to a group
Comment
Footprint
Columns: Name Alias
4.135
Address1
Address2
Address3
Address4
Application_BuildNumb...
ApprovedBy
Author
Capacitor Type
CheckedBy
Class I
Class II

Export... OK Cancel

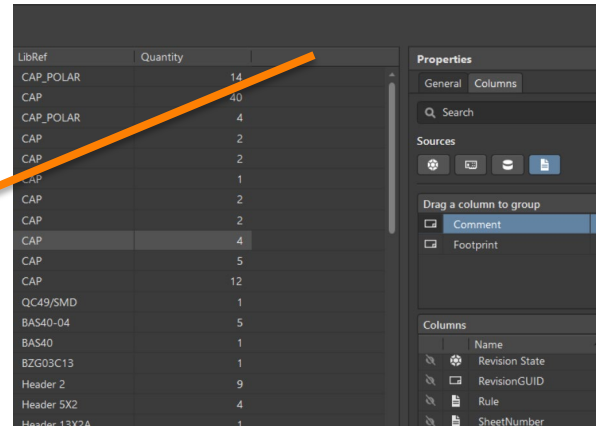
BILL OF MATERIALS

Differences between a project with a component link to a company database, that include actualized supplier and a list without possibilities of include external information from an automated form.



The screenshot shows a BOM table with columns: LibRef, Quantity, and Supplier. The 'Supplier' column contains values like '[NoValue]' and '[NoParam]'. An orange arrow points from the 'Supplier' column header to the 'Supplier' property in the 'Columns' list of the 'Properties' panel on the right.

LibRef	Quantity	Supplier
CAP_POLAR	14	[NoValue]
CAP	40	[NoParam]
CAP_POLAR	4	
CAP	2	
CAP	2	
CAP	1	
CAP	2	
CAP	2	
CAP	4	
CAP	5	
CAP	12	
QC49/SMD	1	
BAS40-04	5	
BAS40	1	
BZG03C13	1	
Header 2	9	
Header 5X2	4	
Header 13X2A	1	
Header 5X2	2	
Header 7X2	1	
Header 10X2	2	
D Connector 9	1	
D Connector 9_MA...	1	
Dsub-15-Female-HD	1	
PS2-6PIN	2	
...	...	...



The screenshot shows a BOM table with columns: LibRef, Quantity, and Supplier. The 'Supplier' column is empty. An orange arrow points from the 'Supplier' column header to the 'Supplier' property in the 'Columns' list of the 'Properties' panel on the right.

LibRef	Quantity	Supplier
CAP_POLAR	14	
CAP	40	
CAP_POLAR	4	
CAP	2	
CAP	2	
CAP	1	
CAP	2	
CAP	2	
CAP	4	
CAP	5	
CAP	12	
QC49/SMD	1	
BAS40-04	5	
BAS40	1	
BZG03C13	1	
Header 2	9	
Header 5X2	4	
Header 13X2A	1	



Agenda

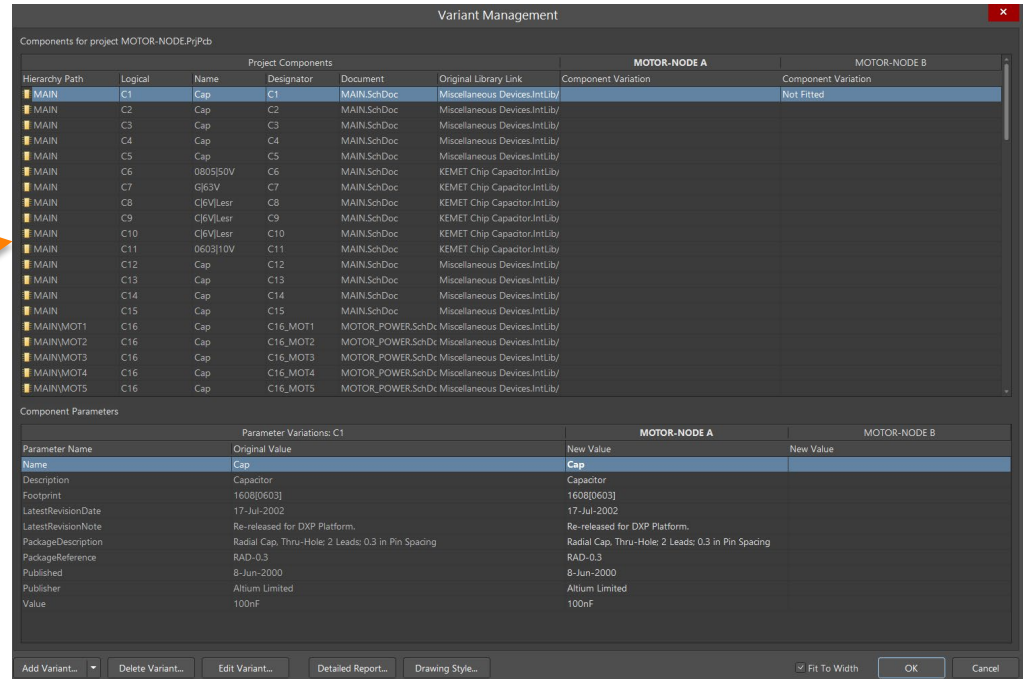
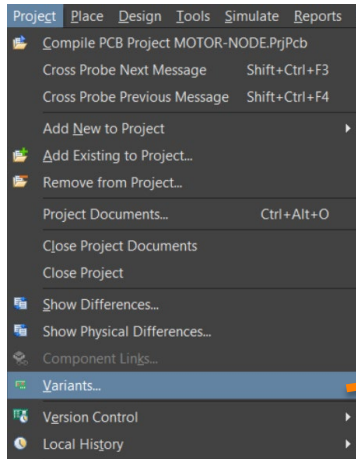
- Harness
- Supplier search
- **Variants**
- SCH pre-layout directives
- Pin Swapping

ASSEMBLY VARIANTS

- Specify 'variants' of the PCB, where the actual layout is the same but there are variations in:
 - The number of original design components it contains
 - The specific part type/value of those components (e.g. a discrete component with a different value, or a faster logic component of the same or different technology family)
 - Varied component-level parameters (e.g. tolerance or voltage-rating of a capacitor, Manufacturer's Part Number, etc).
- There is no need to create different projects, these variants are automatically managed by Altium

ASSEMBLY VARIANTS

The use of variants in the industry is commonly used for manage different device models, with the same logic design.

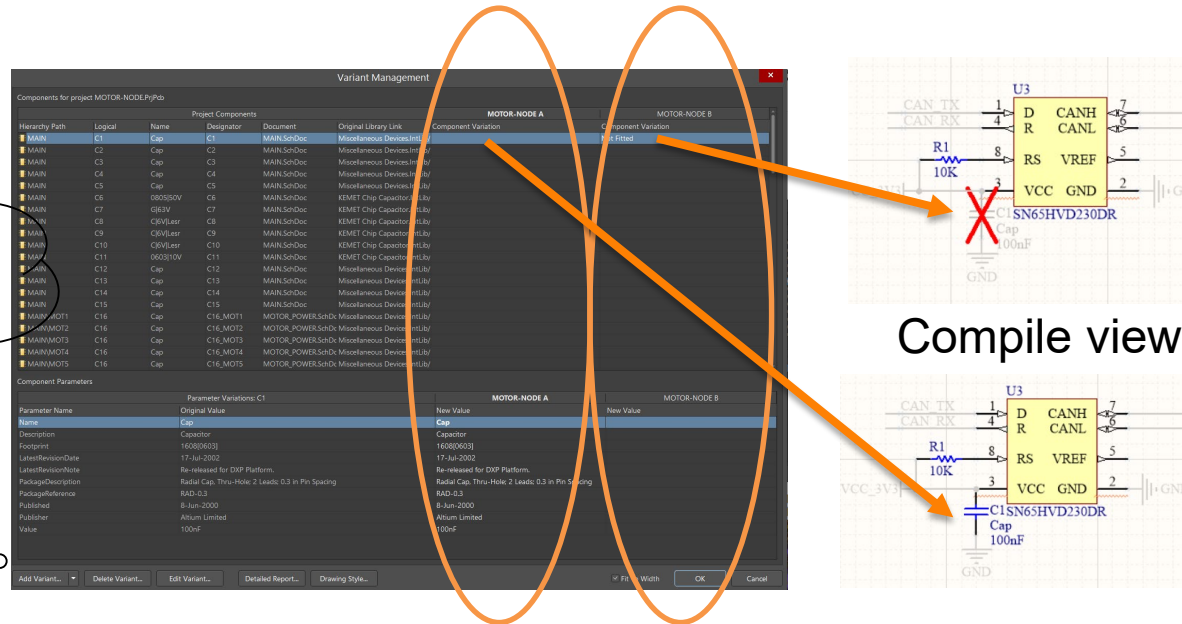


In the Variant Management window we can see all the variants inside the project

ASSEMBLY VARIANTS

On a variant colum, we have to deselected the components that aren't necessary for that release.

With Add button we can include all variants that our project need



ASSEMBLY VARIANTS

We can obtain webform with differences between all variations in the project, or all components.

Altium Designer

Variant Report

Date: 27/08/2018
Time: 7:12:10
Sheet: 000000

Component Name

Original Value

MOTOR-NODE A

MOTOR-NODE B

Physical

Document

Parameter Name

Original Value

MOTOR-NODE A

MOTOR-NODE B

C1

MAIN.SchDoc

Cap

Fitted

Cap

Not Fitted

Comment

Cap

Cap

Description

Capacitor

Capacitor

Footprint

1608(0603)

1608(0603)

LatestRevisionDate

17-Jul-2002

17-Jul-2002

LatestRevisionNote

Re-released for DXP Platform.

Re-released for DXP Platform.

PackageDescription

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

PackageReference

RAD-0.3

RAD-0.3

Published

8-Jun-2000

8-Jun-2000

Publisher

Altium Limited

Altium Limited

Value

100uF

100uF

C2

MAIN.SchDoc

Cap

Fitted

Cap

Fitted

Comment

Cap

Cap

Description

Capacitor

Capacitor

Footprint

0805

0805

LatestRevisionDate

17-Jul-2002

17-Jul-2002

LatestRevisionNote

Re-released for DXP Platform.

Re-released for DXP Platform.

PackageDescription

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

PackageReference

RAD-0.3

RAD-0.3

Published

8-Jun-2000

8-Jun-2000

Publisher

Altium Limited

Altium Limited

Value

100uF

100uF

C3

MAIN.SchDoc

Cap

Fitted

Cap

Fitted

Comment

Cap

Cap

Description

Capacitor

Capacitor

Footprint

0805

0805

LatestRevisionDate

17-Jul-2002

17-Jul-2002

LatestRevisionNote

Re-released for DXP Platform.

Re-released for DXP Platform.

PackageDescription

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

PackageReference

RAD-0.3

RAD-0.3

Published

8-Jun-2000

8-Jun-2000

Publisher

Altium Limited

Altium Limited

Value

100uF

100uF

Variant Management

Components for project MOTOR-NODE.PyJob

Project Components

MOTOR-NODE A

Component Variation

Component Value

Not Fitted

Hierarchy Path

Logical

Name

Designator

Document

Original Library Link

Component Variation

Component Value

Not Fitted

MAIN

C1

Cap

C1

MAIN.SchDoc

Miscellaneous Devices.IntLib

Not Fitted

MAIN

C2

Cap

C2

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN

C3

Cap

C3

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN

C4

Cap

C4

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN

C5

Cap

C5

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN

C6

0805(050V

C6

MAIN.SchDoc

KEMET Chip Capacitor.IntLib

MAIN

C7

GR3V

C7

MAIN.SchDoc

KEMET Chip Capacitor.IntLib

MAIN

C8

QV1Vler

C8

MAIN.SchDoc

KEMET Chip Capacitor.IntLib

MAIN

C9

QV1Vler

C9

MAIN.SchDoc

KEMET Chip Capacitor.IntLib

MAIN

C10

QV1Vler

C10

MAIN.SchDoc

KEMET Chip Capacitor.IntLib

MAIN

C11

0603(10V

C11

MAIN.SchDoc

KEMET Chip Capacitor.IntLib

MAIN

C12

Cap

C12

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN

C13

Cap

C13

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN

C14

Cap

C14

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN

C15

Cap

C15

MAIN.SchDoc

Miscellaneous Devices.IntLib

MAIN/MOT1

C16

Cap

C16_MOT1

MOTOR_POWER.SchDoc

Miscellaneous Devices.IntLib

MAIN/MOT2

C16

Cap

C16_MOT2

MOTOR_POWER.SchDoc

Miscellaneous Devices.IntLib

MAIN/MOT3

C16

Cap

C16_MOT3

MOTOR_POWER.SchDoc

Miscellaneous Devices.IntLib

MAIN/MOT4

C16

Cap

C16_MOT4

MOTOR_POWER.SchDoc

Miscellaneous Devices.IntLib

MAIN/MOT5

C16

Cap

C16_MOT5

MOTOR_POWER.SchDoc

Miscellaneous Devices.IntLib

Component Parameters

Parameter Variations: C1

MOTOR-NODE A

None

Parameter Name

Original Value

New Value

None

Name

Cap

Cap

None

Description

Capacitor

Capacitor

None

Footprint

1608(0603)

1608(0603)

None

LatestRevisionDate

17-Jul-2002

17-Jul-2002

None

LatestRevisionNote

Re-released for DXP Platform.

Re-released for DXP Platform.

None

PackageDescription

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

Radial Cap, Thru-Hole 2 Leads: 0.3 in Pin Spacing

None

PackageReference

RAD-0.3

RAD-0.3

None

Published

8-Jun-2000

8-Jun-2000

None

Publisher

Altium Limited

Altium Limited

None

Value

100uF

100uF

None

Add Variant...

Delete Variant...

Edit Variant...

Detailed Report...

Drawing Style...

Fit To Width

OK

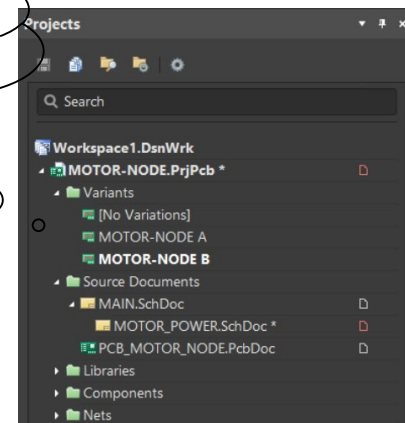
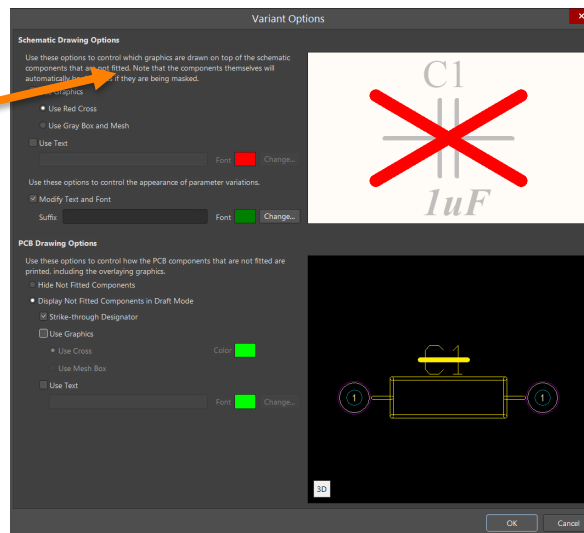
Cancel

ASSEMBLY VARIANTS

On projects pannel we can work with differents variants created by us

Variant Management									
Components for project MOTOR_NODE0.jyLib									
Project Components				MOTOR_NODE A			MOTOR_NODE B		
Hierarchy Path	Symbol	Name	Designator	Document	Original Library Link	Component Variation	Component Variation		
MAIN	C1	Cap	C1	MAIN.SchDoc	Miscellaneous Devices.IntLib		Not fitted		
MAIN	C2	Cap	C2	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN	C3	Cap	C3	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN	C4	Cap	C4	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN	C5	Cap	C5	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN	C6	800520V	C6	MAIN.SchDoc	KEMET Chip Capacitor.IntLib				
MAIN	C7	090V	C7	MAIN.SchDoc	KEMET Chip Capacitor.IntLib				
MAIN	C8	09V16uF	C8	MAIN.SchDoc	KEMET Chip Capacitor.IntLib				
MAIN	C9	09V16uF	C9	MAIN.SchDoc	KEMET Chip Capacitor.IntLib				
MAIN	C10	09V16uF	C10	MAIN.SchDoc	KEMET Chip Capacitor.IntLib				
MAIN	C11	800520V	C11	MAIN.SchDoc	KEMET Chip Capacitor.IntLib				
MAIN	C12	Cap	C12	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN	C13	Cap	C13	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN	C14	Cap	C14	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN	C15	Cap	C15	MAIN.SchDoc	Miscellaneous Devices.IntLib				
MAIN/MOT1	C16	Cap	C16, MOT1	MOTOR_POWER.SchDoc	Miscellaneous Devices.IntLib				
MAIN/MOT2	C16	Cap	C16, MOT2	MOTOR_POWER.SchDoc	Miscellaneous Devices.IntLib				
MAIN/MOT3	C16	Cap	C16, MOT3	MOTOR_POWER.SchDoc	Miscellaneous Devices.IntLib				
MAIN/MOT4	C16	Cap	C16, MOT4	MOTOR_POWER.SchDoc	Miscellaneous Devices.IntLib				
MAIN/MOT5	C16	Cap	C16, MOT5	MOTOR_POWER.SchDoc	Miscellaneous Devices.IntLib				

In the Variant Options window we can define how output files represents the not include elements





Agenda

- Harness
- Supplier search
- Variants
- **SCH pre-layout directives**
- Pin Swapping

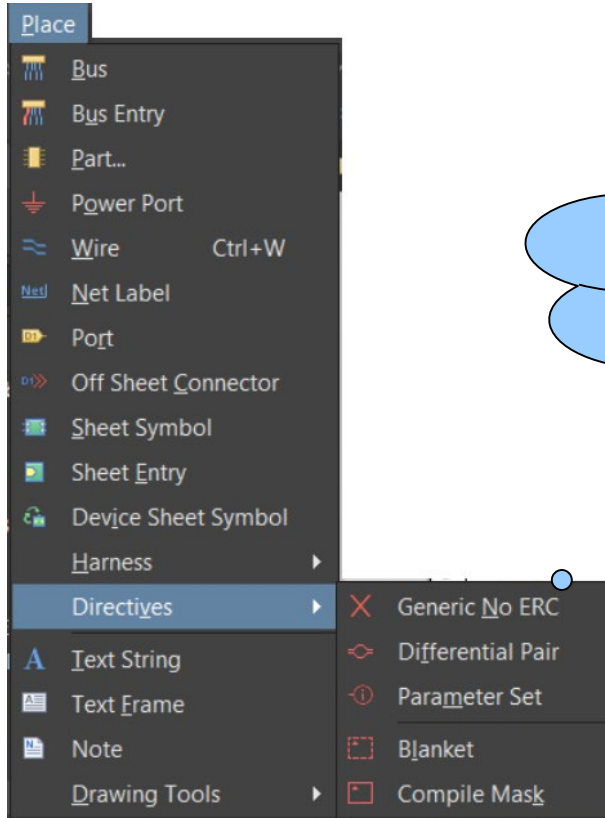
SCH pre-layout directives

There are typically two scenarios in which pre-layout definition is desired:

- You may wish to fully define the design requirements on the schematic, such that the schematic becomes the 'master record' for the design.
- You may have two different people working on the design – one capturing it and the other laying it out on the PCB. The person capturing the design may wish to ensure that the particular constraints they have in mind are indeed used during the layout phase



SCH pre-layout directives



For certain schematic components, we can add relevant parameter(s) as part of that object's properties

The following information can be specified, using:

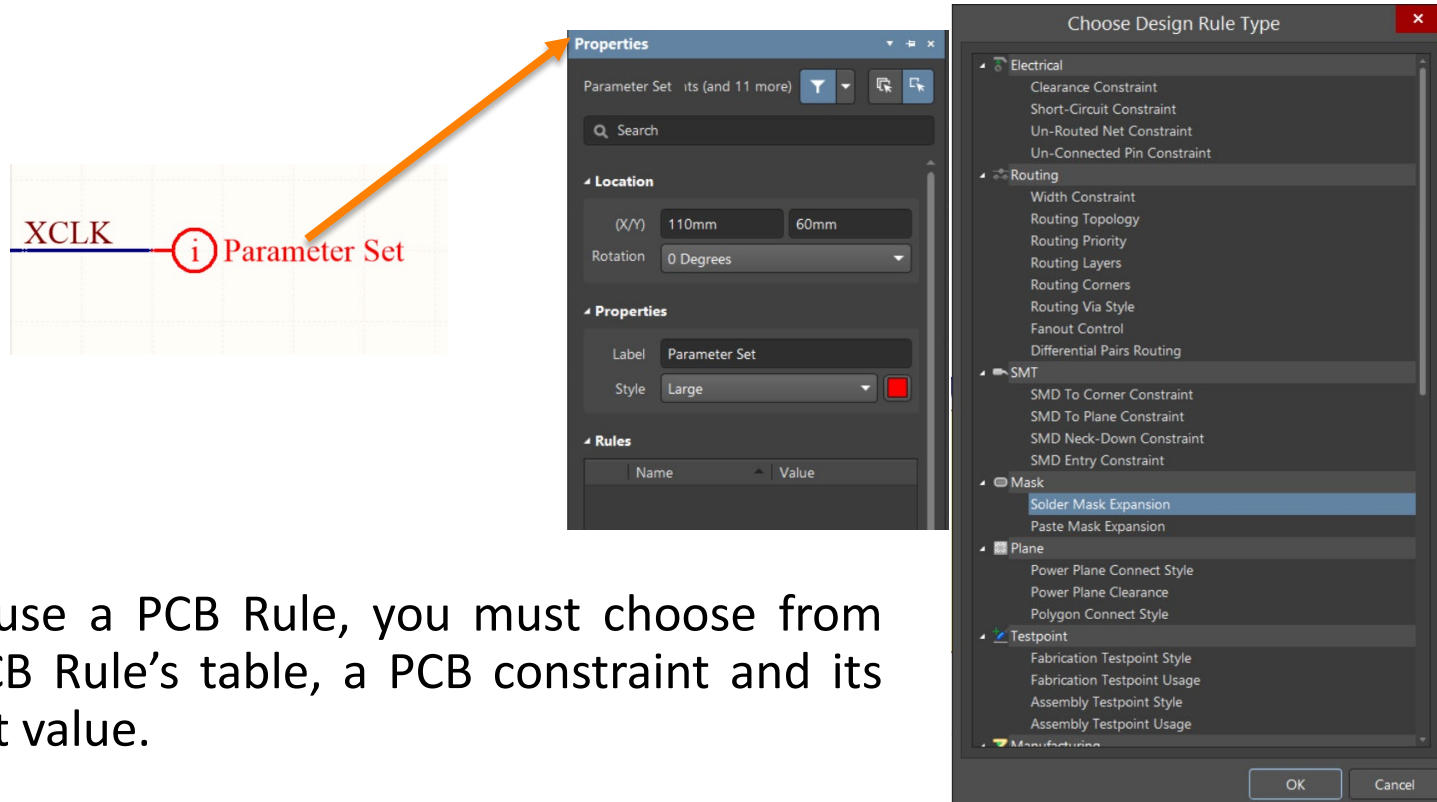
- PCB layout constraints
- Net classes
- Differential pairs

SCH pre-layout directives

What schematic parameter-to-PCB rule scope options are supported on Altium?

Add a Parameter (as a rule) to a...	From...	For a PCB rule scope of...
Pin	the Parameters tab of the <i>Pin Properties</i> dialog	Pad
Port	the Parameters tab of the <i>Port Properties</i> dialog	Net
Wire	the <i>Parameters</i> dialog, after placing a PCB Layout Directive (Parameter Set object) on the wire using the Place » Directives » PCB Layout command	Net
Bus	the <i>Parameters</i> dialog, after placing a PCB Layout Directive (Parameter Set object) on the bus using the Place » Directives » PCB Layout command	Net Class
Harness	the <i>Parameters</i> dialog, after placing a PCB Layout Directive (Parameter Set object) on the harness using the Place » Directives » PCB Layout command	Net Class
Blanket	the <i>Parameters</i> dialog, after placing a PCB Layout Directive (Parameter Set object) on the edge of the Blanket using the Place » Directives » PCB Layout command. Include a <code>ClassName</code> parameter to create a net class for all nets covered by the blanket, which will then be used for the rule scope.	Net Class
Component	the Parameters region of the <i>Component Properties</i> dialog	Component
Sheet Symbol	the Parameters tab of the <i>Sheet Symbol</i> dialog	Component Class
Sheet	the Parameters tab of the <i>Document Options</i> dialog (Design » Document Options)	All Objects

SCH pre-layout directives: parameter set

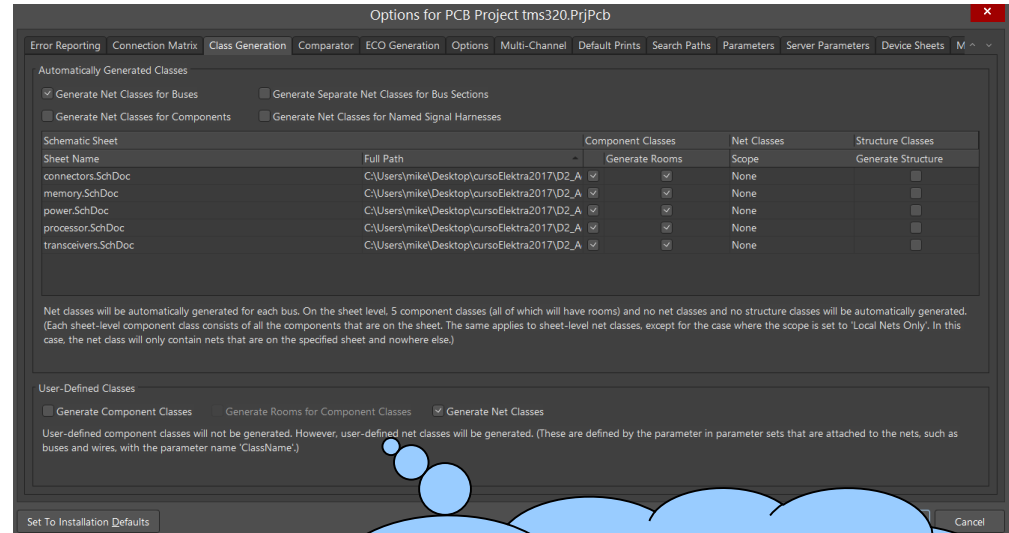


If we use a PCB Rule, you must choose from the PCB Rule's table, a PCB constraint and its correct value.

SCH pre-layout directives

The normal use of Altium does not include the generation of net classes (only in diff. pair nets).

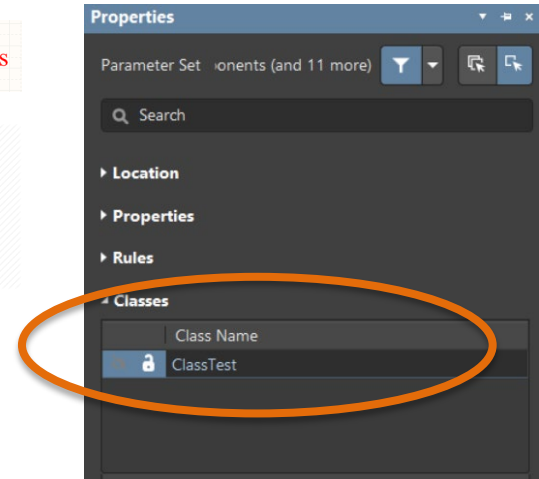
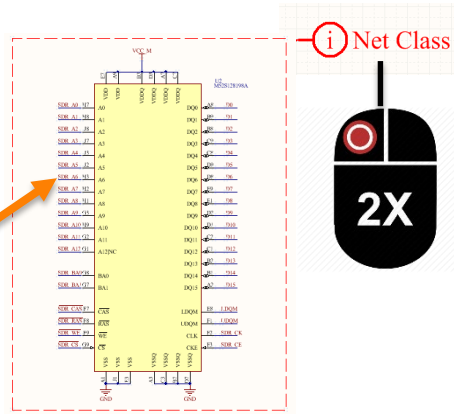
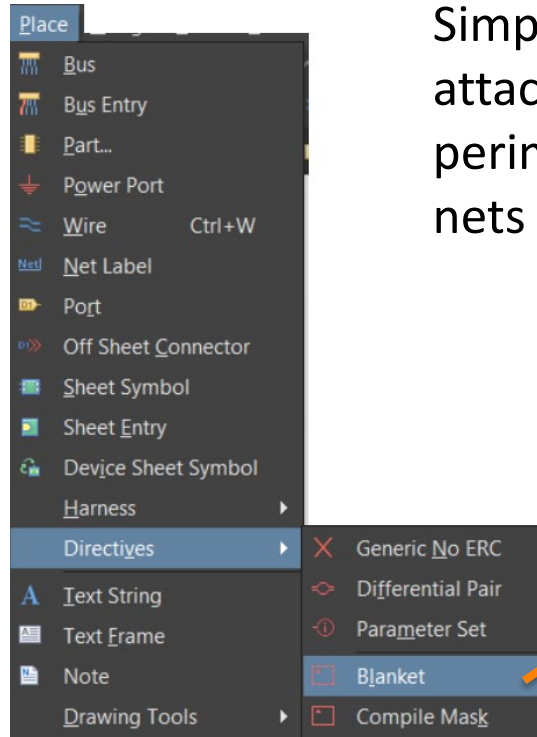
This work is usually made in the PCB workspace (Design>Classes)



To insert Net Class directives we need to enable Class Generation options

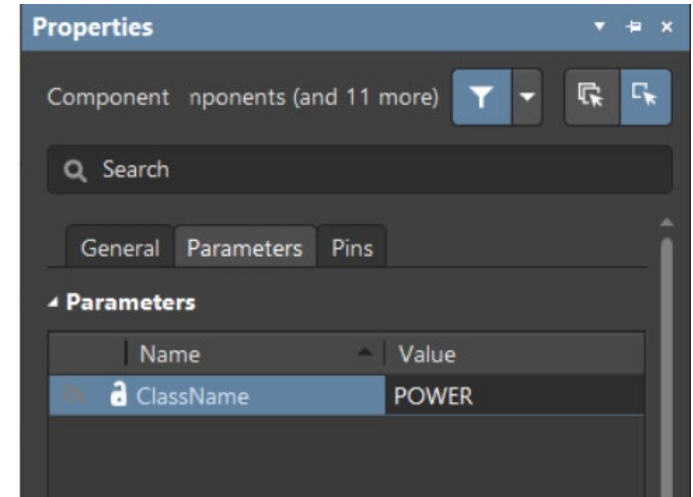
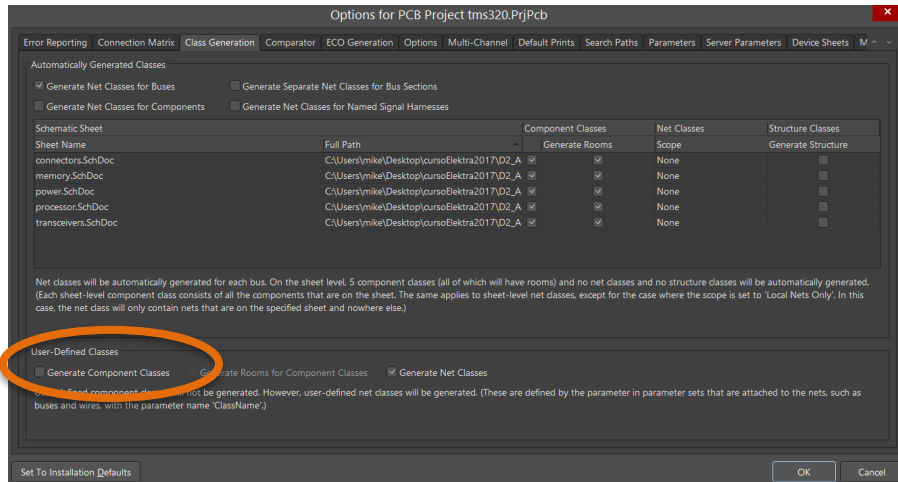
SCH pre-layout directives: blanket

Simply place the Blanket around the required nets, then attach and define a Net Class directive to the blanket's perimeter. The Net Class directive will be applied to all nets that fall within the area defined by the blanket.



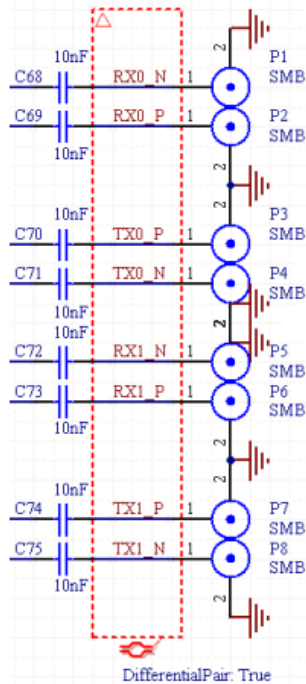
SCH pre-layout directives

User-defined component classes are handled in the schematic through the "ClassName" parameter in the symbol's properties.



Adding a new parameter called "ClassName" and change the value of the parameter to your desired component class name that you gave them in the PCB.

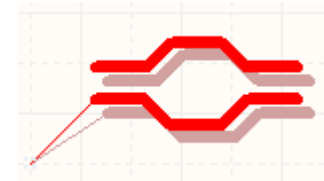
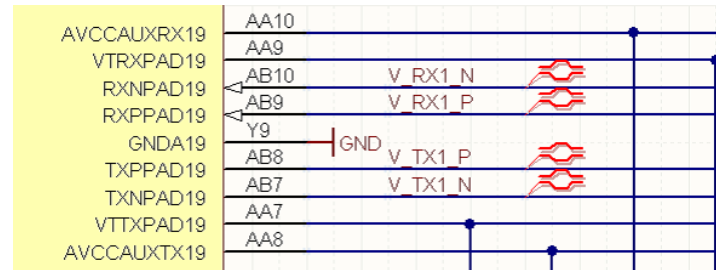
SCH pre-layout directives: diff pair



Differential pairs are defined on the schematic, we have to use two ways for placing them:

- A differential pair directive (Place » Directive) on each of the nets in the pair.
- A blanket area for a group of differential pairs. Inserting only one pair directive for all group.

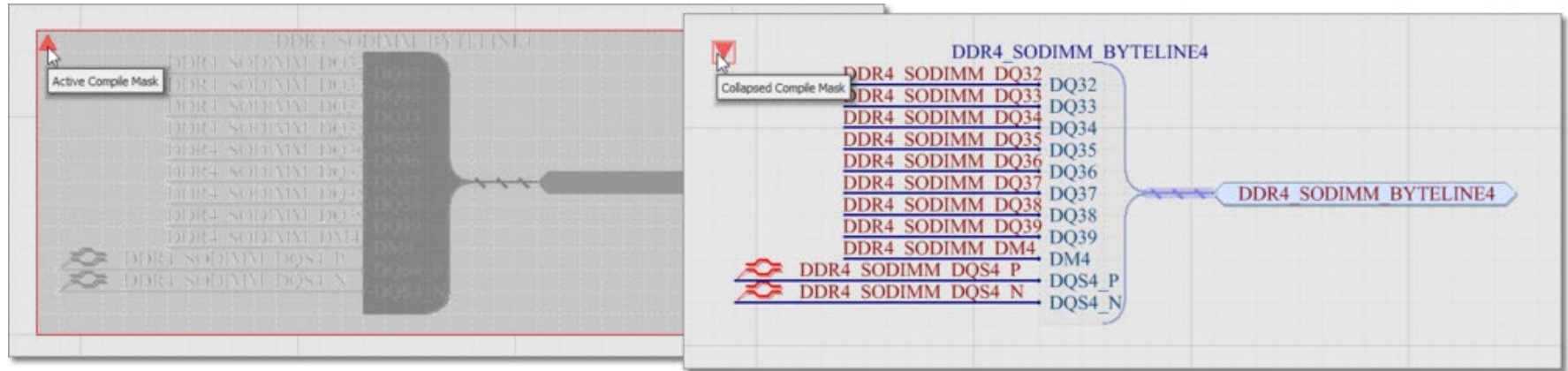
The net pair must be named with net label suffixes of _N and _P.



SCH pre-layout directives: compile mask



A Compile Mask is a design directive that is used to effectively hide the area of the design it contains from the Compiler. It provides a way to manually prevent error checking for circuitry that may not yet be complete and will generated compile errors.



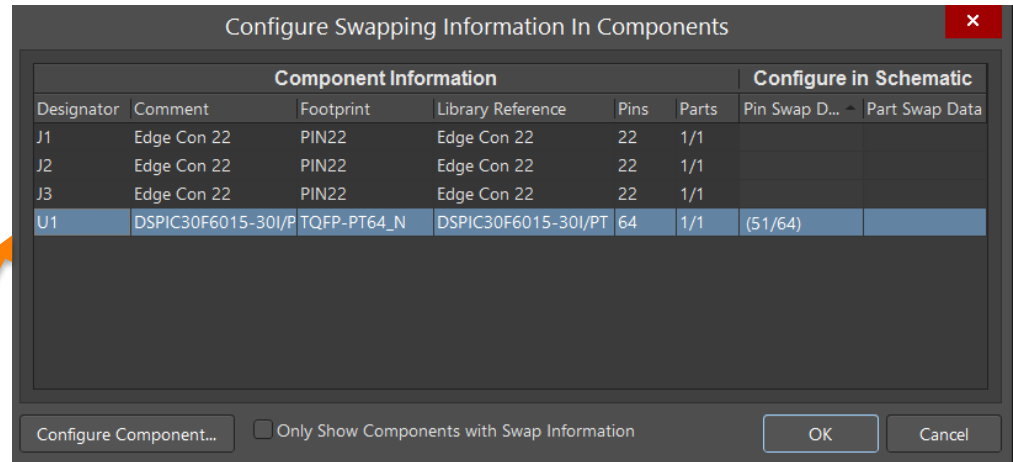
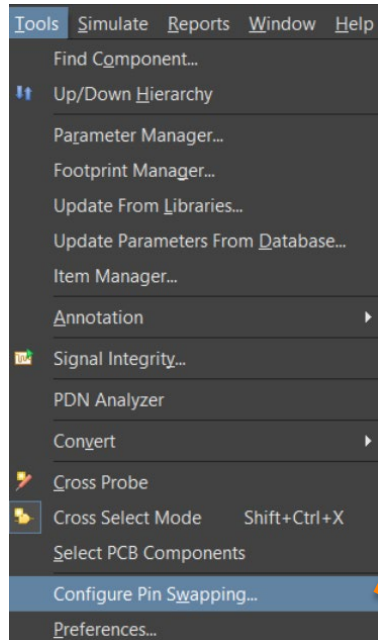


Agenda

- Harness
- Supplier search
- Variants
- SCH pre-layout directives
- **Pin Swapping**

PIN/PART SWAPPING

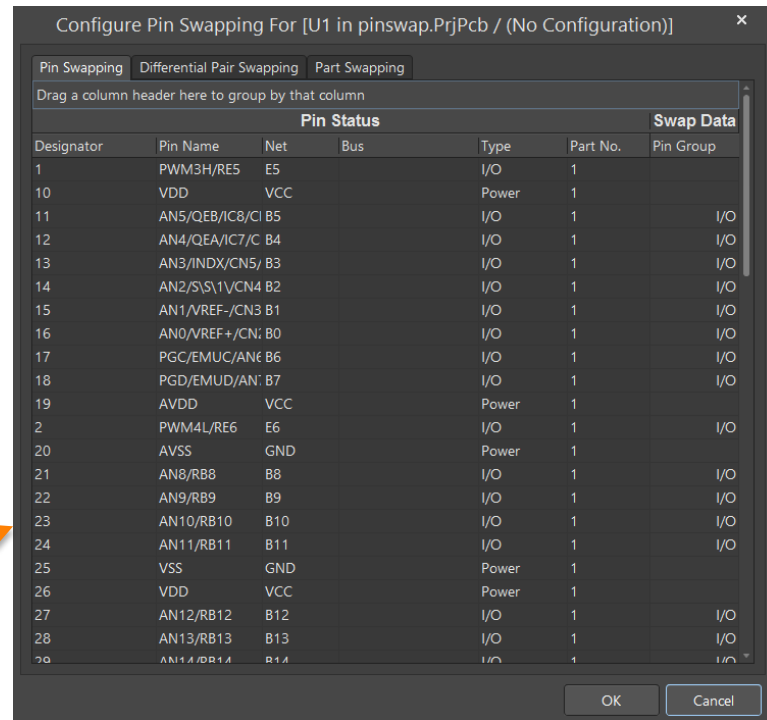
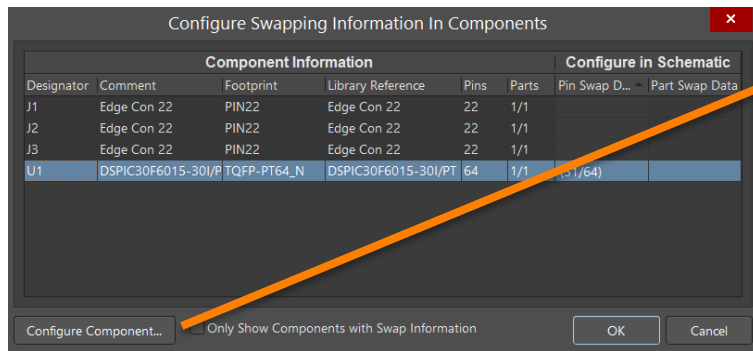
This feature provides all the benefits of traditional pin/part-swapping systems. During a pin swap operation Altium Designer analyses the net assigned to the chosen pin, and dynamically reassigns the net on the pin and any connected copper.



PIN/PART SWAPPING

A component pin is swappable with another pin in that component when it belongs to the same *pin group* (has the same *pin group* value).

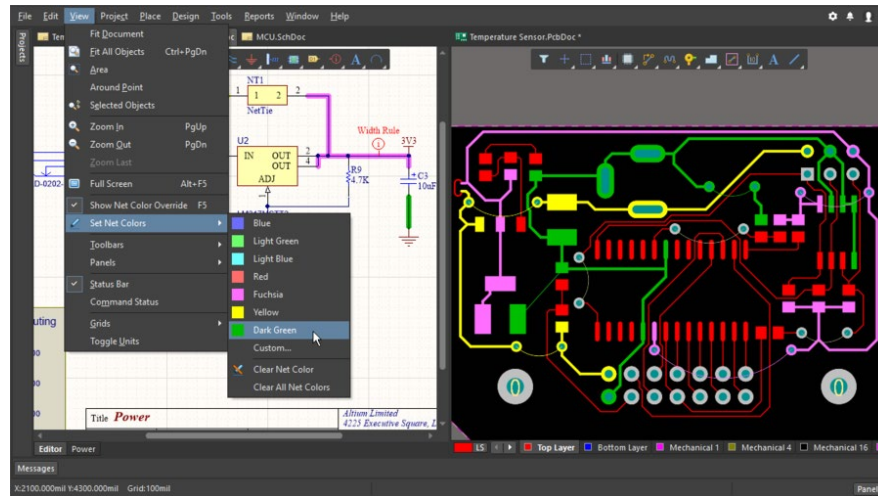
The *pin group* is an attribute of each pin in the component and its value can be any alphanumeric string.



NET COLOURS

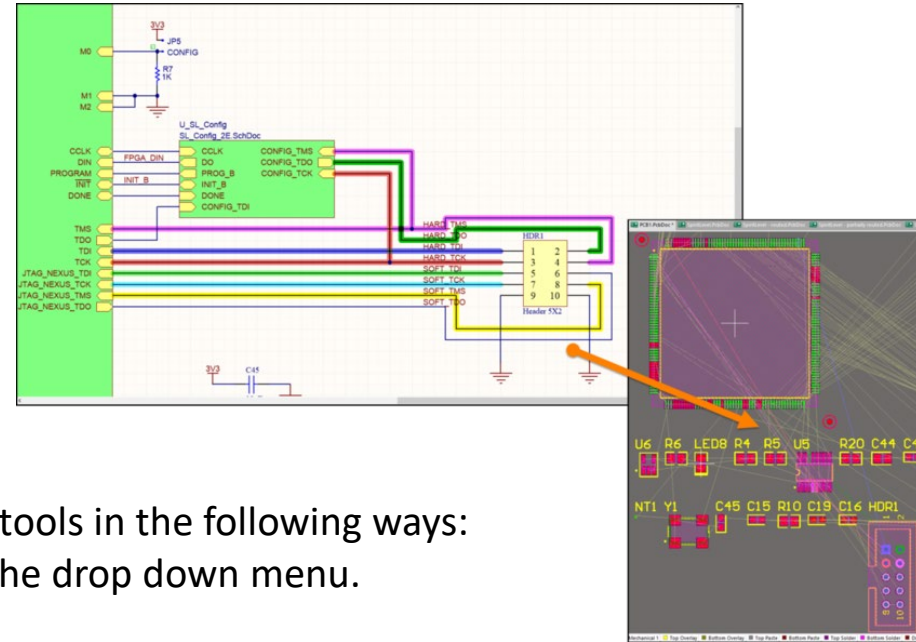
As the design builds up it quickly becomes dense and detailed. To help bring clarity and visibility, the designer can apply colour to the schematic wiring.

Colour can be applied to schematic nets and those colour settings transferred to the PCB, or colour settings can be applied directly in the PCB editor



NET COLOURS

We can highlight multiple nets, differentiate different nets with different colours, and push highlighting from schematic to PCB and vice versa. The highlighting feature now assigns colour to the entire net in the schematic instead of only coloring the selected wire.



In the schematic, you can access the net highlighting tools in the following ways:

- From the toolbar, click  and select a color from the drop down menu.
- From the menu, select **View » Set Net Colors**.
 - In the PCB document **Net Color Override** must be selected in the PCB for these colours changes to be visible by selecting **View » Net Color Override** from the toolbar



Agenda

END OF THE MODULE